# Highlights

**Hierarchical relations guide memory retrieval in sentence comprehension: Evidence from a local anaphor in Turkish**

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- Three visual world studies examined antecedent retrieval for the Turkish reciprocal.
- Stimuli disentangled syntactic hierarchical relations from clause, case and subjecthood, and linear order.
- Hierarchically available targets were rapidly distinguished from distractors.
- There was limited evidence for interference from distractors.
- Hierarchical relations between NPs determine their availability in memory.

# Hierarchical relations guide memory retrieval in sentence comprehension: Evidence from a local anaphor in Turkish

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**Abstract**

Real-time sentence comprehension relies on memory resources to establish syntactic dependencies. Such dependency formation is subject to hierarchical constraints on interpretation. However, it remains an open question how hierarchical information is represented and used in working memory. Here we asked whether hierarchical relations between noun phrases as in *x c-commands y* (Reinhart, 1976) constrain antecedent retrieval for a local anaphor. We measured the real-time reactivation of hierarchically available target antecedents and unavailable distractors in processing the Turkish reciprocal *birbirleri* via three visual world studies. Our design deconfounded hierarchical relational information between noun phrases from other structural information (clause-mateness, case, subjecthood) and linear order/recency. Experiment 1 compared the reactivation of hierarchically available subjects and clause-mate, similarly case-marked, hierarchically unavailable distractors. Subjects were rapidly distinguished from distractors within the reciprocal window, irrespective of their linear order. Experiment 2 revealed that hierarchically available indirect objects showed a similar advantage at retrieval. Experiment 3 was a pre-registered, high-powered replication of this result. Overall, we found that hierarchical relations between noun phrases within a single clause rapidly determine antecedent availability in retrieval. Our findings suggest that either linguistic representations encode hierarchically informed features targeted by retrieval cues, or that hierarchical information shapes the organization of linguistic representations in memory.

*Keywords:* sentence processing, memory retrieval, relations, syntax, Turkish

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## 1. Introduction

Hierarchical structure has long been considered the hallmark of human language syntax (Chomsky, 1957; Joshi et al., 1975; Kaplan and Bresnan, 1982; Steedman, 2000). Accordingly, it is thought to play a fundamental role in language comprehension by guiding the incremental interpretations that comprehenders assign to sentences, and has been the central focus of many theories of real-time language use (e.g., Dillon, 2011; Frazier and Clifton Jr., 1996; Gibson, 1998, 2000; Keshev et al., 2025; Kush, 2013; Levy, 2008; Phillips et al., 2011; Sturt, 2003; Wagers, 2008; cf. Frank and Bod, 2011).

Hierarchical structure constrains the interpretation of long-distance dependencies that may span multiple intervening words. For example, in (1) the anaphor *each other* must be interpreted in relation to its antecedent *kids*. Despite the presence of feature-matching (i.e., plural) referents in the sentence, i.e., *seniors* and *neighbors*, *kids* is the only available antecedent. This is due to its hierarchical position relative to the anaphor within the sentence's syntax, expressed in (for example) Principle A of Binding Theory (Chomsky, 1981; Reinhart, 1976): *Kids* is local (e.g., clause-mate) to the anaphor and hierarchically prominent (e.g., in a c-commanding position) relative to it within the embedded clause.

(1) *The seniors*$_{\mathrm{i}}$ mentioned that the *neighbors*$_{\mathrm{j}}$' <u>kids</u>$_{\mathrm{k}}$ playfully teased **each other**$_{*\mathrm{i}/*\mathrm{j}/\mathrm{k}}$.

Such hierarchical constraints are commonly defined in terms of structural relationships between pairs of linguistic items within a sentence, as in *x is local to y* or *x c-commands y*.

One question for theories of real-time dependency formation is how to integrate such relational constraints between items with theories of the psychological mechanisms that subserve real-time dependency formation (Alcocer and Phillips, 2012; Dillon, 2011; Kush, 2013; Wagers, 2008). On the one hand, there is an abundance of evidence that comprehenders activate numerous interpretations of linguistic input rapidly and in parallel (e.g., MacDonald et al., 1994). Such a view would suggest that any grammatically licensed interpretation of the input could be made available very early in the comprehension process.

On the other hand, theories of dependency formation often invoke a content-based, associative retrieval mechanism (e.g., Lewis and Vasishth, 2005; Lewis et al., 2006; McElree, 2000, 2006; Van Dyke and McElree, 2011). On this view, linguistic items are encoded in short term memory with their *item-specific* features, including structural (e.g., subjecthood, case) and morphological (e.g., number, gender) information. Dependency formation requires reactivating these encodings on demand in response to a set of retrieval cues. Dependency formation should be rapid when cued by the features of a target item, as this allows rapid retrieval from working memory (Lewis and Vasishth, 2005; Lewis et al., 2006; McElree et al., 2003). But relational notions such as c-command or relative linear order are less straightforwardly encoded in such an architecture (Alcocer and Phillips, 2012; Dillon and Keshev, 2024; Kush, 2013; Lewis and Vasishth, 2005; Wagers, 2008). If comprehenders must infer or reconstruct the relationship between two elements to establish that they are in an appropriate configurational relationship (Dillon, 2011; Dillon et al., 2014; Kush et al., 2015; Lewis and Vasishth, 2005), then dependency formation could be slower or less reliable (Lewis and Vasishth, 2005; McElree and Dosher, 1993; Ratcliff and McKoon, 1989; cf. Dosher, 1983).

In an attempt to address this question, much research has investigated the role of structural relations in the processing of anaphors, addressing their time-course and fidelity. This body of work has shown that comprehenders rapidly employ structural constraints to link anaphors with their antecedents. But it does not definitively establish whether comprehenders specifically use hierarchical or relational constraints to do so (Dillon et al., 2013; Kush and Phillips, 2014; Kush et al., 2015). This work has also examined whether structurally unavailable antecedents can interfere in dependency formation. In the present study, we focus primarily on the question of whether hierarchical information guides dependency formation in the earliest stages of comprehension, largely setting aside the question of interference from structurally unavailable distractors.

More specifically, we ask: Does real-time antecedent retrieval rely on hierarchical relations between noun phrases (NPs) as in *x c-commands y*? We report three visual world studies that measured the moment-by-moment reactivation of potential antecedents in processing a local anaphor in Turkish. We employed stimuli carefully designed to isolate the role of c-command relation between NPs in a sentence from other sources information. We achieved this by manipulating the hierarchical c-command relations of the potential antecedents with respect to the anaphor while holding their clause relations and case marking constant. Across our studies, we tested targets in c-commanding subject and object positions. Furthermore, we manipulated linear order and feature-match of antecedents with respect to the anaphor to reveal the impact of recency and feature-based interference on retrieval, respectively. To preview our results, we found that the abstract hierarchical relation between NPs in a sentence determines antecedent availability above and beyond other sources of information from the earliest moments of processing an anaphor. The finding that abstract hierarchical relations immediately guide interpretation constrains psycholinguistic theories of dependency formation, which we take up in the discussion.

### 1.1. *The role of hierarchical relations in real-time processing*

The processing of local anaphors such as reflexives (*himself, herself, themselves*) and reciprocals (*each other*) has been a focus of research because they are common linguistic elements that are subject to relatively well understood syntactic relational constraints on interpretation such as clause-mateness and *c-command*. The c-command constraint is defined on the basis of the hierarchical organization of individual noun phrases (NPs) in the syntactic configuration of a sentence. Formally, a node $\alpha$ c(onstituent)-commands a node $\beta$ if $\beta$ is $\alpha$'s sister or is contained within $\alpha$'s sister in the tree structure (Büring, 2005; Reinhart, 1976, 1983). For example, in the sentence *The seniors mentioned that the neighbors' kids playfully teased each other* (repeated from (1) above), the target subject *kids* c-commands the anaphor *each other* because the anaphor is contained within the verb phrase (VP) that is the sister of the target subject's NP, while *neighbors* does not c-command the anaphor as it is embedded within this subject's NP as a possessor and is not a sister to the same VP[1].

Another way to formulate the hierarchical relational constraint between a local anaphor and its antecedent is via coargumenthood, which is a central concept in certain theories for anaphor licensing (Pollard and Sag, 1994; Reinhart and Reuland, 1993). Because target antecedents of local anaphors are typically both in a c-commanding position and coarguments of the anaphor, the two factors often cooccur, making it difficult to definitively isolate c-command. Nonetheless, we frame the rest of the discussion here as well as the current study with reference to c-command as it is a fundamental notion governing the distribution of many syntactic dependencies (but see the General Discussion).

Many empirical studies on the processing of anaphors have shown that comprehenders rapidly distinguish hierarchically available, e.g., c-commanding and clausemate, targets from distractors, which is consistent with the use of relational information in early memory processes (Dillon et al., 2013, 2014; Han et al., 2021; Kush

[1]We adopt the notion of c-command to characterize the abstract geometric relations within constituent structure that constrain anaphoric dependencies. We acknowledge that in addition to c-command, a range of other proposals have been advanced in the literature to identify similar relational constraints within a configurational hierarchy including *command* (Langacker, 1969), *kommand* (Lasnik, 1976), *maximal-projection-command* (Chomsky, 1981) and *precede-and-command* (Bruening, 2014). All of these proposals converge on the core idea that a hierarchical relational constraint between an anaphor and its antecedent is necessary to describe anaphoric dependencies; in this paper we use the notion of c-command as it is widely recognized in syntactic theory.

et al., 2015; Runner and Head, 2014; Sturt, 2003; Xiang et al., 2009). However, these findings do not necessarily implicate hierarchical, relational processing, as individual studies typically admit multiple interpretations. For example, Dillon et al. (2014) argues that comprehenders use structural information of 'clausemateness' to constrain the initial stages of memory retrieval when processing the Mandarin Chinese anaphor *ziji* (see also, Dillon et al., 2016). While it is possible to consider 'clausemate' a relational constraint, this is not a logical necessity. Clause level information could be encoded as an item-level feature, which may have independent psychological motivation if a clause is considered a single temporal context (Howard and Kahana, 2002) in the form of events anchored by its verb (Wagers, 2008). This would allow a simple, feature-based encoding of clausal context that can be cued in retrieval (Balachandran, 2024; Polyn et al., 2009; Sederberg et al., 2008; Wagers, 2008).

More compellingly, several studies have tried to isolate the role of c-command in processing bound variable pronouns (Cunnings et al., 2015; Kush et al., 2015; Moulton and Han, 2018). For example, Kush et al. (2015) compare the processing of bound variable and coreferential pronouns. They report an eye-tracking-while-reading experiment with stimuli as in (2) crossing antecedent type, i.e., quantificational (*any janitor*) or referential (*the janitor*), and the c-command relation between the pronoun *he* and the antecedent using either *when* or *but* clauses. In (2-a) the bound variable reading for *he* is available with the *when* clause because the *when* clause is contained within the embedded verb phrase of *liked*, allowing *he* to be c-commanded by *any janitor*. The bound variable reading is unavailable with the *but* clause because the *but* clause is combined with the matrix clause via a conjunction phrase higher up, not allowing *he* to be c-commanded by *any janitor*. A coreferential interpretation between *he* and *the janitor* in (2-b), on the other hand, is available irrespective of the type of the clause. If antecedent retrieval for bound variable pronouns uses hierarchical relational information such as the c-command constraint, extra processing difficulty is expected only when the quantificational NP *any janitor* does not c-command the pronoun *he* in a *but* clause.

(2) a. Kathi didn't think *any janitor* liked performing his custodial duties, when/but *he* had to clean up messes left after prom anyway.
b. Kathi didn't think *the janitor* liked performing his custodial duties, when/but *he* had to clean up messes left after prom anyway.

The results reveal no evidence for a difference between the two conditions with a referential NP, but increased difficulty in reading the post-pronoun region when *any janitor* does not c-command *he* (i.e., in the *but* clause condition). In a follow-up experiment, they report that feature match of quantificational NPs in a non-c-commanding position did not affect early reading measures, with only suggestive evidence for delayed interference effects from gender-matching, non-c-commanding quantificational NPs in later measures (for similar findings, see Cunnings et al., 2015). In a later study, Moulton and Han (2018) disentangle the syntactic notion of c-command from the semantic notion of scope, finding that it was the syntactic c-command relation that determined antecedent availability in processing bound variable pronouns. Overall, these results show that comprehenders use hierarchical relational processing to distinguish available from unavailable quantificational NPs when interpreting bound variable pronouns.

Still, it's possible that Kush et al.'s (and Cunnings et al.'s and Moulton and Han's) findings are driven by the hierarchical position of an entire clause (e.g., *when* vs. *but* clauses in Kush et al.) relative to the critical antecedent in a different clause. That is, the structural manipulation in those studies concerns items across different clauses. Thus, although their findings do provide evidence for the use of clausal relations in structuring memory retrieval, they do not show evidence for relational processing within a single clause.

There is also evidence that the hierarchical organization of NPs impacts the availability of their memory representations for other types of processing. For example, testing subject-verb agreement cases in a probe recognition task, Franck and Wagers (2020) show that attractor nouns that c-command the verb in the surface order are more readily available in memory than those that do not (for similar findings in judgment tasks, see Franck et al., 2006, 2010, 2020). However, such findings do not necessarily indicate that c-commanding nouns are represented or accessed from memory on the basis of relational processing. Rather, as discussed by Franck and Wagers, these findings may reflect increased activation levels for c-commanding nouns in memory.

### *1.2. A review of local anaphor processing: A gap in the role of hierarchical structure*

To date, dozens of studies have explored the real-time processing of anaphors, potentially establishing them as one of the most commonly studied dependencies in psycholinguistic research. On the surface, there appears to be a considerable amount of evidence that memory retrieval in real-time processing of anaphoric dependencies is sensitive to hierarchical information (e.g., Dillon et al., 2013; Nicol and Swinney, 1989; Sturt, 2003; Xiang et al., 2009). Studies employing different methodologies show that comprehenders rapidly differentiate the hierarchically available *target* antecedent from the hierarchically unavailable *distractor*. This is evidenced by the priming of semantic associates of targets in cross-modal priming tasks (Nicol and Swinney, 1989), increased difficulty in reading time measures in the absence of a feature-matched target antecedent (e.g., Dillon et al., 2013; Sturt, 2003), increased positivities in event-related potentials upon processing an anaphor with a feature-mismatching target (e.g., Xiang et al., 2009), and increased looks to targets over distractors in visual world studies (e.g., Han et al., 2021; Runner and Head, 2014). However, while these results provide clear evidence that comprehenders do use some type of structural information in retrieval, they do not unambiguously show that a hierarchical or relational notion like c-command *per se* is used to accomplish this.

Against the rich empirical background on anaphoric dependencies, it is perhaps surprising that the role of hierarchical relations in resolving anaphoric items remains unclear. But one key observation about existing studies is that most used stimuli that confounded multiple sources of structural information in defining the target of retrieval, failing to uniquely isolate the role of c-command or similar hierarchical notions in retrieval. Instead, a range of structural, item-level information could be used to retrieve the target antecedent instead of a relational notion like c-command. To substantiate this observation, we conducted a review of the literature on processing local anaphors, summarized in the table in Appendix A. Our crosslinguistic review includes 45 experiments from 28 studies on local anaphors that included structurally unavailable distractors. The studies are predominantly in English, with some in Chinese, German, Hebrew, Hindi, Korean, Russian, Swedish, Tagalog and Turkish. We examined the experimental stimuli in these studies to determine what information was used to distinguish the target and distractor. Our review also includes details about the studies (e.g., language under investigation, number of participants) and summarizes the findings as originally presented.

We find that no study in our review directly isolates the role of c-command *per se* without confounding it with other potential structural cues such as clause-mateness and/or case marking. Importantly, these are cues that are plausibly construed as item-level features, obviating the need to implement hierarchical or relational structure at retrieval. In our analyses of the stimuli from previous studies we use *case marking* to refer to the grammatical role of a noun phrase in the sentence or its morphophonological marking, as we consider both types of case marking to be plausibly relevant—and both can be construed as item-level features. Accordingly, we consider the target and distractor to be differently case-marked when they have either distinct (abstract) structural case marking, as this points to differing grammatical roles in the sentence (e.g., subject vs. object)[2], or different overt morphophonological case

[2]We note that the claim that abstract case marking directly modulates processing difficulty is not a shared view. Abstract-case-associated effects are sometimes modeled based on the resulting argument roles (e.g., subjecthood) rather than the abstract case feature itself (e.g., Patil et al.,

marking.

A summary of our review is illustrated with a Venn diagram in Figure 1. The diagram shows the structural/relational cues (i.e., *C-command*, *Clause*, *Case*) that distinguish the target from the distractor.

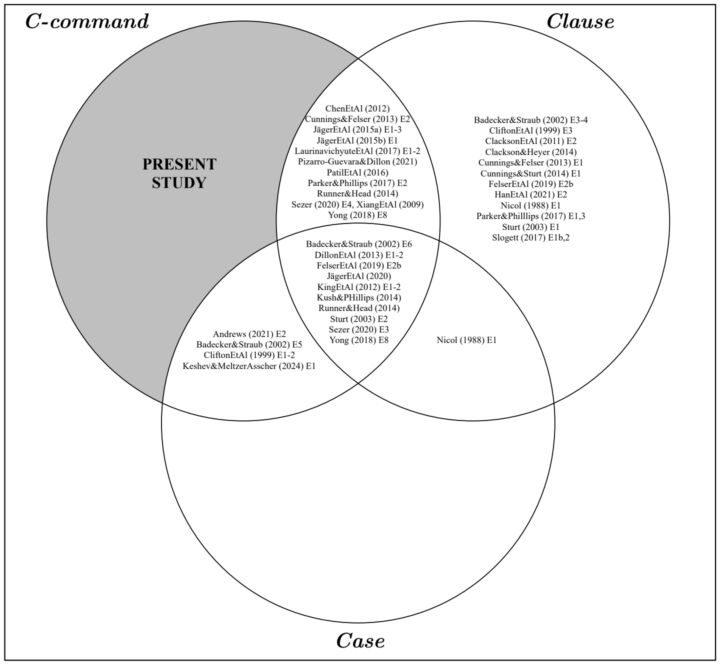


Figure 1: Venn diagram illustrating the structural/relational information manipulated in previous studies on local anaphor processing to distinguish structurally available targets from structurally unavailable distractors.

To illustrate further, 12 experiments from 10 studies create a distractor by manipulating all three structural/relational features. An example is (3-a). The distractor (*middle manager*) is in a non-c-commanding position, in a different clause from the target (*new executive*), and has a different case (i.e., grammatical role) from the target. Fifteen experiments from 12 studies have a non-c-commanding, non-clause-mate distractor (e.g., *Fred* in (3-b)). Five experiments from 4 studies have a non-c-commanding, differently-case-marked distractor (e.g., *fireman* in (3-c)). One study has a non-clause-mate, differently-case-marked distractor, (e.g., *janitor* in (3-d)). Lastly, 15 experiments from 12 studies have a non-clause-mate distractor (e.g., *John*

2016).

in (3-e)). In (3) anaphors are given in bold, target antecedents are underlined and distractors are in italic.

(3) a. The new executive$_{\mathrm{i}}$ [who oversaw *the middle manager*$_{\mathrm{j}}$] apparently doubted **himself**$_{\mathrm{i/*j}}$ on most major decisions.
b. The tough soldier$_{\mathrm{i}}$ [that *Fred*$_{\mathrm{j}}$ treated in the military hospital] introduced **himself**$_{\mathrm{i/*j}}$ to all the nurses.
c. The son$_{\mathrm{i}}$ of *the fireman*$_{\mathrm{j}}$ hurt **himself**$_{\mathrm{i/*j}}$ in a bad accident.
d. *The landlord*$_{\mathrm{i}}$ told *the janitor*$_{\mathrm{j}}$ [that the fireman$_{\mathrm{k}}$ with the gas-mask would protect **himself**$_{\mathrm{*i/*j/k}}$ if it became necessary].
e. *John*$_{\mathrm{i}}$ thought [that Bill$_{\mathrm{j}}$ owed **himself**$_{\mathrm{*i/j}}$ another opportunity to solve the problem].

As discussed above, clause-mateness could be driven by item-level features indexing temporal/contextual information (Balachandran, 2024; Wagers, 2008). Likewise, in all the studies that we review the target antecedent was the subject of the clause containing the anaphor. Comprehenders could thus rely on a composite, item-level cue such as *subject-of-the-current-clause* in accessing the target (e.g., Arnett and Wagers, 2017; Kush and Phillips, 2014; Schoknecht et al., 2025; Xiang et al., 2009). In either case, retrieval could succeed without appealing to hierarchical or relational processing among elements within a clause.

### *1.3. The current study*

While many acknowledge the importance of testing relational notions like c-command in retrieval, our review shows that it has proven difficult to uniquely isolate their role in the processing of anaphors. Our study here aims to fill this gap by testing whether hierarchical relations between NPs within a clause can be used to guide antecedent retrieval. We investigate the preverbal, local reciprocal *birbirleri* in Turkish, which must be bound by a c-commanding, local (e.g., clause-mate) antecedent[3] (Akkuş, 2021; Kornfilt, 1997). Turkish provides an ideal empirical ground to test this: (i) Its rich morphological system allows us to manipulate the hierarchical availability of antecedents while eliminating other confounding cues (i.e., clause, case and subjecthood), and (ii) its head-final structure allows us to investigate a preverbal anaphor, which is crucial for eliminating any confounds related to reactivation of

[3] *Birbirleri* can also be bound by a non-c-commanding or non-local antecedent (see the examples in Kornfilt, 1997; Paparounas and Akkuş, 2024). However, these are often limited to certain syntactic or semantic environments, and are not relevant to the local binding conditions that we examine in this study.

other arguments at the verb (for a discussion of the importance of this point, see Dillon et al., 2013; King et al., 2012).

We examine the availability of c-commanding subjects (Subj) in Experiment 1 (as simplified in (4-a)) and c-commanding indirect objects (IOs) in Experiment 2 (as in (4-b)), both in contrast to non-c-commanding distractors located within the same embedded clause as the anaphor. Experiment 3 is a high-power, pre-registered replication of the core contrasts in Experiments 1 and 2 (as in (4-c)). We control for case marking and linear order and manipulate the number (mis-)match between the anaphor and the critical referents to test the advantage of c-commanding antecedents above and beyond these factors.

(4) a. *Experiment 1*
Subj [Subj-PL-GEN distractor(-PL)-GEN **anaphor** verb] verb
Subj [distractor(-PL)-GEN Subj-PL-GEN **anaphor** verb] verb
b. *Experiment 2*
Subj [Subj(-PL)-GEN IO(-PL)-DAT **anaphor** verb] verb
Subj [Subj-PL-GEN distractor-(-PL)-DAT **anaphor** verb] verb
c. *Experiment 3*
Subj [Subj-PL-GEN IO-/distractor-PL-DAT **anaphor** verb] verb
Subj [Subj-PL-GEN distractor-PL-GEN **anaphor** verb] verb

We use the visual world paradigm for spoken language comprehension in our experiments (Cooper, 1974; Tanenhaus et al., 1995). This choice contrasts with many studies on processing anaphors, which generally employ reading methodologies. However, reading measures only generate indirect inferences about memory retrieval processes through variations in reading times (e.g., slow-downs or speed-ups). In contrast, the visual world paradigm offers a more direct window into the competition between the target antecedents and distractors by tracking overt looks to their visual referents, as there is a well-documented relationship between fixation behavior and attention to the referents of visually presented objects (Allopenna et al., 1998; Magnuson, 2019). It is also highly suitable for investigation of real-time anaphor resolution due to its high temporal resolution (e.g., Runner et al., 2003; Han et al., 2021).

## 2. Data Availability

The data, analysis scripts, experimental stimuli, and example recordings for the present study are available from https://osf.io/7g3z8.

## 3. Experiment 1: C-command and recency

Experiment 1 investigates the processing of the Turkish local reciprocal *birbirleri*. We designed stimuli to isolate the role of c-command from other potential cues such as clause-mateness, case marking, topicality, linear order/recency and feature-match.

We predicted that if the c-command relation is rapidly used to guide antecedent retrieval, then there would be more looks to c-commanding targets than distractors across all conditions immediately within the reciprocal region. We further predicted that if a cue-based retrieval mechanism is used to retrieve antecedents, this effect may be further moderated by number feature match and/or recency: More recent, feature matching distractors should be more likely to be erroneously retrieved at the critical anaphor (e.g., Jäger et al., 2015b, 2017; Parker and Phillips, 2017; Sturt, 2003).

### *3.1. Materials*

The stimuli in Experiment 1 are as in (5). The critical anaphor is the direct object of an embedded verb, and the target antecedent is the subject of that embedded clause. The distractor is a subordinated noun in the same embedded clause, i.e., embedded either in the subject NP as in (5-a) or in an adjunct NP as in (5-b). In hierarchical terms, the phrase of the target subject is a sister to the node containing the anaphor; hence, it c-commands the anaphor. However, the phrase of the distractor is not a sister to the same node; hence, it does not c-command the anaphor. The stimuli are in a 2×2 within-subject design, crossing RECENCY of the target subject with respect to the distractor, i.e., Recent Target as in (5-a) and Distant Target as in (5-b), and number MATCH between the plural anaphor and the plural or singular distractor, i.e., Distractor Match and Distractor Mismatch, respectively. In all conditions, the only available antecedent that can bind *birbirleri* is the local, c-commanding referent, i.e., *kameramanlar* 'cameramen'.

(5) Example item from Experiment 1 (with anaphor in bold, target underlined and distractor in italic)

a. Recent Target, Distractor Match / Mismatch

| | | | | |
|---|---|---|---|---|
| Ayşe$_i$ | [*yönetmen(-ler)-in*$_j$ | yeni | <u>kameraman-ların-ın</u>$_k$ | sertçe |
| Ayşe | director-PL-GEN | new | cameraman-3PL.POSS-GEN | severely |

| | | |
|---|---|---|
| **birbirlerin-i**$_{*i/*j/k}$ | eleştir-dik-lerin]-i | anlat-tı. |
| each.other-ACC | criticize-NMLZ-3PL.POSS-ACC | say-PST |

'Ayşe$_i$ said that the director(s)'s$_j$ new cameramen$_k$ severely criticized each other$_{*i/*j/k}$.'

b. Distant Target, Distractor Match / Mismatch

| | | |
|---|---|---|
| Ayşe$_i$ | [kameraman-lar-ın$_j$ | *yönetmen(-ler)-in*$_k$ |
| Ayşe | cameraman-PL-GEN | director-PL-GEN |

| | | |
|---|---|---|
| izn-i-yle | sertçe | **birbirlerin-i**$_{*i/j/*k}$ |
| permission-3SG.POSS-COM | severely | each.other-ACC |

| | |
|---|---|
| eleştir-dik-lerin]-i | anlat-tı. |
| criticize-NMLZ-3PL.POSS-ACC | say-PST |

'Ayşe$_i$ said that the cameramen$_j$, with the director(s)'s$_k$ permission, severely criticized each other$_{*i/j/*k}$.'

In all conditions, both the target and the distractor are clausemates to the anaphor and marked with the genitive case -*(n)In*. Genitive case has different functions on the target and the distractor: On the target, it marks the subject of an embedded clause; on the distractor, it marks the possessor noun in a genitive-possessive construction.

We factorially manipulate the linear order/recency of the target relative to the distractor by using two different structures, as shown in (5). To keep the saliency of the target and distractor comparable, we introduce a third referent as the discourse topic, i.e., *Ayşe*, in the matrix subject position.

Twenty-four sets of the four experimental conditions above were created and distributed across four lists in a Latin Square design. Before the visual world study, an offline antecedent selection task verified that *birbirleri* must be bound by the clausemate, c-commanding target antecedent, and that non-c-commanding distractors are not available as antecedents in the experimental stimuli, with each selected 98% and 14% of the time, respectively (for details, see Appendix B.1).

In addition to the experimental items, the study included 72 fillers that mirrored the structure of the experimental trials but replaced the anaphor with a name referent. This referent was either the distractor in the sentence (e.g., *yönetmen(ler)* 'director(s)' in (5)) in one-third of the fillers, the plural unmentioned referent in another third, or the singular unmentioned referent in the remaining third. This distribution ensured that each image had an equal likelihood of being mentioned just prior to the critical anaphor onset, preventing participants from predicting the referent in advance.

The auditory stimuli were recorded by the first author, who is a native speaker of Turkish. Prior to the study, all visual stimuli were normed for codability (for further details, see Appendix C).

### *3.2. Participants*

Sixty-eight native speakers of Turkish participated in the study ($M_{age} = 22$). They were undergraduate or graduate students at a public university in Istanbul, Turkey, and were recruited through social media. They had normal or corrected-to-normal vision. Each of them received 200 Turkish liras (∼$10.5 at the time of the study) for compensation. Data from two participants were removed as they had track loss in 18 and 22 of their experimental trials, leaving 66 participants for analyses.

### *3.3. Procedure*

Before the experiment, the participants were informed about the procedure and asked to fill out a written consent form and a brief demographic survey. Participants were tested individually in a private testing booth in a laboratory located at a public university in Istanbul, Turkey. Eye-movements were recorded via a desktop eye tracker, SR Eye Link 1000 Plus (SR Research, Ontario, Canada), with a sampling rate of 1000 Hz. Although viewing was binocular, only data from the right eye was recorded. A chin rest minimized head movements. Each session started with a calibration process, and further calibrations were done throughout the experiment as needed. There were three practice trials to familiarize the participants with the task. All instructions were given in Turkish. Each session took about an hour, and the participants were offered three breaks throughout the experiment.

In each trial in the study, the participants first listened to a lead-in sentence, then a question, and lastly a critical sentence through headphones. An example lead-in sentence and question for the example critical sentence in (5) are in (6) below. The lead-in sentence provided background for the corresponding experimental sentence, and the question cued the participants to click on the referent of the reciprocal that would appear in the following sentence. The lead-in sentence included the nominal form of the embedded verb in the experimental sentence. The question had the embedded verb itself in passivized form to allow participants to anticipate it in the experimental sentence; and it targeted the noun bearing the thematic role of the anaphor (e.g., the people who were *criticized*). Questions were given *before* the experimental sentence to direct participants' eye movements towards the referent of the anaphor as quickly as possible. This goal-oriented design allowed us to infer referential decisions from eye-movements during the processing of critical anaphor (Trueswell, 2008).

(6) a. Lead-in

| | | | | | |
|---|---|---|---|---|---|
| Ayşe | maalesef | birçok | eleştiri-ye | şahit | ol-du. |
| Ayşe | unfortunately | several | criticism-DAT | witness-PST | |

'Ayşe witnessed several criticisms unfortunately.'

b. Question

Peki Ayşe kim-in eleştir-il-diğ-in-i anlat-tı?
so Ayşe who-GEN criticize-PASS-NMLZ-3SG.POSS-ACC say-PST
Tıkla-yın.
click-IMP
'So who were the people that Ayşe said were criticized? Click.'

The screen was blank during the presentation of the lead-in sentence and the question. At the end of the question, a preview of the visual display corresponding to the critical sentence was presented, schematized in Figure 2. This structure aimed to restrict eye movements driven by the visual properties of the images to the preview session as much as possible (Henderson and Ferreira, 2004). The visual display consisted of four images: the target and the distractor from the experimental sentence, and a singular and a plural referent that were snot mentioned in the respective trial. The participants' task was to click on the image of the referent that answered the preceding question. No feedback was given. The participants' eye-movements were only recorded during the presentation of the experimental sentences.

### *3.4. Data analyses*

The data were preprocessed using the VWPre package in R (Porretta et al., 2022). All on-screen looks that did not fall on any of the four interest regions were recorded as outside. A dataset that was binned for time slices of 5 ms was created for plots and analyses.

The main time window of interest for the analyses was the *reciprocal*, and its onset in each item set was treated as time zero in the analyses. An additional time window was the following word; the embedded *verb*. The duration of the two windows was the same across the conditions of an experimental item, but differed across items because of different case morphology on the reciprocal and lexical variation of the verb. The average durations of the reciprocal and verb in Experiment 1 were 649 and 782 ms, respectively. In the analyses, both time windows began at 150 ms post-stimulus onset to account for the delay in launching a saccadic eye movement in response to a stimulus (Rayner et al., 1983).

Two analyses for each critical time window were conducted. First, we analyzed gaze proportions to measure competition between the target vs. distractor. However, since this analysis could be influenced by precritical looking behavior, we also analyzed new fixations towards the target and the distractor that were launched within

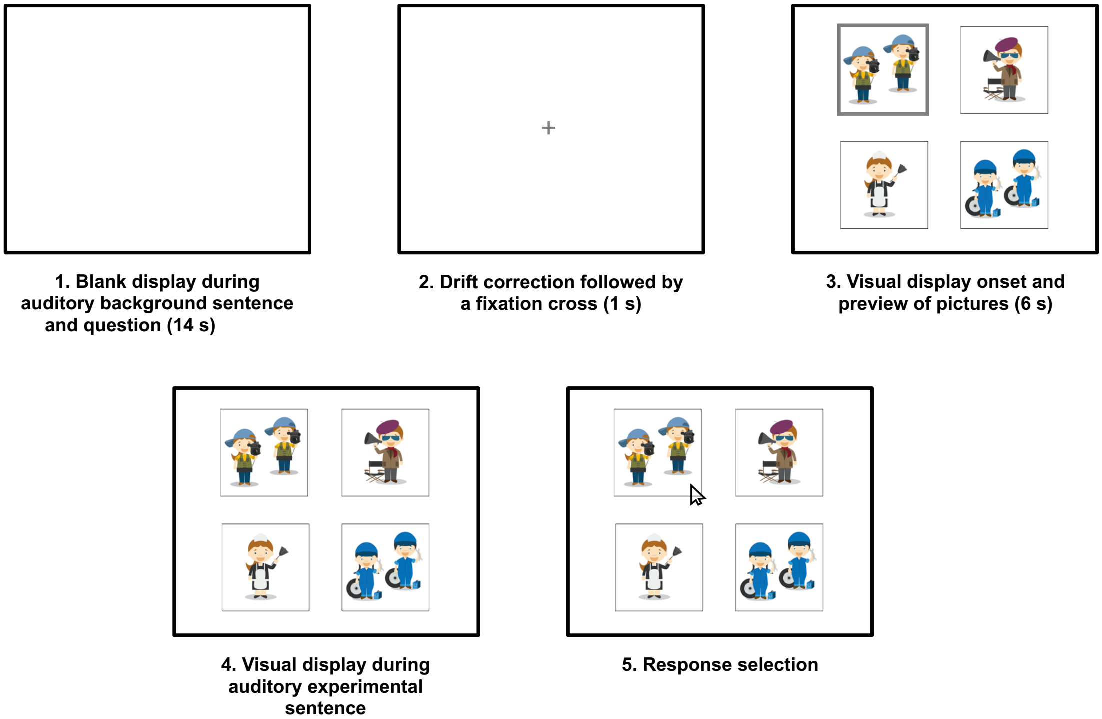


Figure 2: Schematization of experimental trials. In this example the visual display is for the stimuli in (5), with the corresponding target and distractor on the top left and right, and the unmentioned referents at the bottom.

the critical time windows.

The data were analyzed in R, using mixed-effects linear or logistic regression models (Baayen et al., 2008). The *lmer* or *glmer* functions of the LME4 package were used (Bates et al., 2015). Fixed effects in Experiment 1 were RECENCY and MATCH, with random effects for participants and items. We fit two statistical models for each set of analyses: a crossed and a nested model. The crossed model included both fixed effects and their interaction, and the nested model had a main effect of RECENCY and separate MATCH contrasts within the Recent Target and Distant Target conditions. Contrast coding for both models is provided in Table 1. Each model included random intercepts by participants and by items, and random slopes when models converged (Barr et al., 2013). We report $t$- or $z$-scores, with an absolute value of 2 or above indicating significance at the $\alpha$ level .05 (Gelman and Hill, 2006).

| | Crossed Model | | | Nested Model | | |
|---|---|---|---|---|---|---|
| | Recency | Match | Recency:Match | Recency | Recent: Match | Distant: Match |
| Recent Target, Distractor Match | 0.5 | 0.5 | 0.25 | 0.5 | 0.5 | 0 |
| Recent Target, Distractor Mismatch | 0.5 | -0.5 | -0.25 | 0.5 | -0.5 | 0 |
| Distant Target, Distractor Match | -0.5 | 0.5 | -0.25 | -0.5 | 0 | 0.5 |
| Distant Target, Distractor Mismatch | -0.5 | -0.5 | 0.25 | -0.5 | 0 | -0.5 |

Table 1: Contrast coding coefficients used for the crossed and nested statistical models in Experiment 1. Experimental conditions are given in rows, model contrasts given in columns.

*3.4.1. Analysis of corrected log gaze proportion ratios*

Analyses of gaze proportions in Experiment 1 were conducted on *corrected log target vs. distractor ratio*, computed separately for the reciprocal and verb windows. This dependent variable compared the total gaze proportion towards the target relative to the distractor that was corrected by the looks to the unmentioned referents. The formula for the variable is given in (7) (for a similar measure, see Arai et al., 2007). In (7) P(target/unmPL) refers to the proportion of looks to the target divided by looks to the unmentioned plural referent, and P(distractor/{unmSG,unmPL}) refers to looks to the distractor divided by looks to the unmentioned referent matching the number of the distractor (singular or plural). The final target vs. distractor ratio was calculated by subtracting the corrected looks to the distractor from those to the target on the log scale.

(7) ln(P(target/unmPL)) - ln(P(distractor/{unmSG,unmPL}))

Looks to the critical referents were normalized by looks to unmentioned referents to reduce the influence of visual competition in gaze patterns. Competitors with similar features can attract eye movements during visual search (Dahan and Tanenhaus, 2005), and in anaphor resolution, looks to the target or distractor may reflect both referential processing and such visual competition effects (Cunnings and Sturt, 2014). Since unmentioned referents are not linguistically relevant, we assumed that looks to those would reflect visual competition only and used them as a baseline to isolate referential effects.

The ratios were calculated for each trial, for the reciprocal and verb windows separately, and then entered into mixed-effects linear regression models.

*3.4.2. New fixation analysis*

In visual world paradigm, the effect of an experimental factor that has begun in previous time windows may persist in a following time window, even if the factor no longer influences eye movements. Analyses on gaze proportion ratios may thus be confounded with effects lingering from pre-critical windows, which may obscure inferences on the timing of effects. Therefore, we also conducted analyses on the proportions of new fixations that were initiated within the two critical time windows (for a similar analysis, see Kingston et al., 2016). We restricted this analysis to trials in which participants initiated a new fixation towards any of the four interest areas on the display. That is, trials with no fixation towards any of the four interest areas within the critical windows were excluded. We also did not count fixations towards an interest area if the relevant trial already started with looks to that area in the relevant window; so the analyses only include *new* fixations.

We conducted two analyses: one examining the proportion of a new fixation on the target, and the other on the distractor, each relative to all new fixations. For each analysis, a binary dependent variable was created. In the former, trials with a new fixation on the target were coded as 1, and those without as 0. In the latter, trials with a new fixation on the distractor were coded as 1, and those without as 0. The two measures were calculated for each trial, for the reciprocal and verb windows separately, and then entered into mixed-effects logistic regression models.

### *3.5. Results*

Participants mean accuracy across filler items was 96.5%. Trials that had a track loss of 50% or more were removed, which resulted in the loss of 0.008% of the data in experimental sentences.

#### *3.5.1. Picture selection data*

Due to an error in coding the experiment, participants' choices in non-target answers were not recorded; so we only report whether or not they chose the target in Table 2. The results show that the target picture was overall chosen around 96.5% of the time.

| | Target % |
|---|---|
| Recent Target, Distractor Match | 100 |
| Recent Target, Distractor Mismatch | 99.7 (0.3) |
| Distant Target, Distractor Match | 90.6 (2.1) |
| Distant Target, Distractor Mismatch | 95.7 (1.5) |

Table 2: Mean percentages (and SEs over by-participant means) for target picture selections in Experiment 1.

A visual inspection of the data reveals a trend towards fewer target choices when it is linearly distant, with an additional decrease in the presence of a plural distractor. Since we cannot analyze the choices fully, we limit our interpretation to observable trends, and do not report inferential statistics.

*3.5.2. Gaze proportion analyses*

Figure 3 plots the mean raw gaze proportions towards the four interest areas in each condition, starting from the onset of the reciprocal and spanning a period of 1500 ms with time slices of 5 ms (for raw gaze proportions for the entire sentence, see Appendix D).

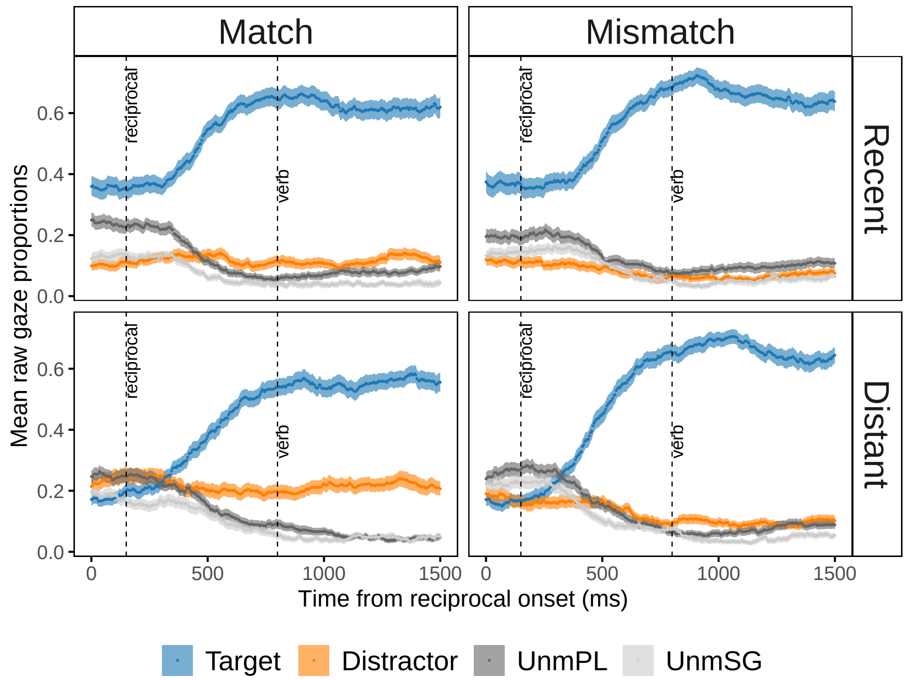


Figure 3: Mean raw gaze proportions towards the four interest areas in Experiment 1. Error bands show SEs over by-participant means. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Proportions are time-locked to the onset of the critical anaphor and span 1500 ms. Dashed lines mark the 150-ms shifted onsets of the reciprocal and verb windows.

Figure 4 shows the corrected log target vs. distractor ratios for the two critical time windows. As observed with positive target vs. distractor ratios in all conditions, there are overall more looks to the target than distractor. Results from the fitted models[4] are in Table 3.

In model output, the intercept is the estimated mean of corrected log target/distractor gaze proportion ratio, averaged across the four experimental conditions. For the re-

[4]Model formulas for all the analyses on the visual world data in Experiments 1-3 are provided in Appendix E.

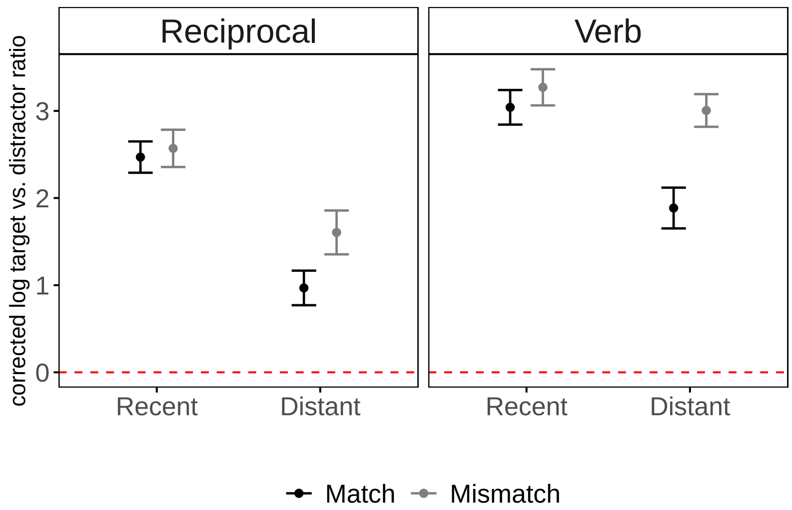


Figure 4: Mean of corrected target vs. distractor ratios on the log scale for the reciprocal and verb windows in Experiment 1. Error bars show SEs over by-participant means. Red dashed lines at 0 indicate equal looks to the target and distractor; values above 0 reflect greater looks to the target.

ciprocal window, the intercept estimate was well above 0 in both models, indicating overall more looks to the target relative to the distractor. The target received overall more looks when it was linearly more recent to the anaphor than the distractor. The nested model showed that in conditions in which the target was more distant, the distractor received more looks when it was plural compared to when it was singular. There was no such difference when the target was more recent.

In the verb window, we again see a clear, and even stronger, overall preference for the target. The target received overall more looks when it was more recent to the anaphor than the distractor. Moreover, plural distractors received overall more looks than singular ones, and there was also a significant interaction between RECENCY and MATCH of the distractor. The nested model illuminates this interaction: When the target was more distant than the distractor, plural distractors received more looks than singular ones. There was no such difference when the target was more recent.

*3.5.3. New fixation analyses*

Figure 5 shows the proportions of new fixations towards the target and distractor out of all new fixations within the reciprocal and verb windows. Corresponding analyses are in Table 3.

In model output, the intercept estimate is the log odds of a new fixation towards

| | Reciprocal | | | Verb | | |
|---|---|---|---|---|---|---|
| | Estimate | Std.error | *t/z* | Estimate | Std.error | *t/z* |
| *Corrected log target vs. distractor ratios* | | | | | | |
| *Crossed model* | | | | | | |
| (Intercept) | **1.90** | **0.19** | **10.27** | **2.80** | **0.17** | **16.48** |
| Recency | **1.23** | **0.22** | **5.62** | **0.71** | **0.19** | **3.73** |
| Match | -0.36 | 0.20 | -1.80 | **-0.68** | **0.18** | **-3.69** |
| Recency:Match | 0.54 | 0.40 | 1.34 | **0.89** | **0.35** | **2.50** |
| *Nested model* | | | | | | |
| (Intercept) | **1.90** | **0.19** | **10.25** | **2.80** | **0.17** | **16.48** |
| Recency | **1.23** | **0.22** | **5.62** | **0.71** | **0.19** | **3.71** |
| Recent: Match | -0.09 | 0.25 | -0.36 | -0.24 | 0.23 | -1.04 |
| Distant: Match | **-0.63** | **0.31** | **-2.05** | **-1.12** | **0.30** | **-3.76** |
| *New fixations towards target* | | | | | | |
| *Crossed model* | | | | | | |
| (Intercept) | **0.89** | **0.10** | **9.34** | **-0.24** | **0.07** | **-3.31** |
| Recency | -0.01 | 0.16 | -0.08 | -0.18 | 0.16 | -1.12 |
| Match | -0.17 | 0.16 | -1.09 | -0.03 | 0.15 | -0.22 |
| Recency:Match | **0.97** | **0.28** | **3.53** | 0.40 | 0.28 | 1.42 |
| *Nested model* | | | | | | |
| (Intercept) | **0.90** | **0.10** | **9.28** | **-0.23** | **0.08** | **-2.77** |
| Recency | 0.03 | 0.16 | 0.21 | -0.18 | 0.16 | -1.13 |
| Recent: Match | 0.33 | 0.25 | 1.32 | 0.15 | 0.21 | 0.68 |
| Distant: Match | **-0.65** | **0.19** | **-3.47** | -0.24 | 0.20 | -1.20 |
| *New fixations towards distractor* | | | | | | |
| *Crossed model* | | | | | | |
| (Intercept) | **-1.27** | **0.10** | **-12.66** | **-0.88** | **0.11** | **-8.36** |
| Recency | -0.26 | 0.14 | -1.80 | **-0.39** | **0.16** | **-2.48** |
| Match | 0.35 | 0.18 | 1.98 | 0.37 | 0.20 | 1.83 |
| Recency:Match | 0.25 | 0.33 | 0.78 | -0.14 | 0.37 | -0.39 |
| *Nested model* | | | | | | |
| (Intercept) | **-1.24** | **0.10** | **-12.88** | **-0.88** | **0.11** | **-8.19** |
| Recency | -0.25 | 0.14 | -1.76 | **-0.37** | **0.16** | **-2.39** |
| Recent: Match | **0.46** | **0.21** | **2.14** | 0.31 | 0.26 | 1.22 |
| Distant: Match | 0.20 | 0.19 | 1.06 | 0.47 | 0.27 | 1.75 |

Table 3: Coefficient estimates, and corresponding standard errors and $t/z$ values for the fixed effects in mixed-effects linear/logistic regression models on corrected log target vs. distractor ratios and new fixations towards the target and distactor in Experiment 1. $|T/z|$-values that are bigger than 2 indicate statistical significance, and are given in bold.

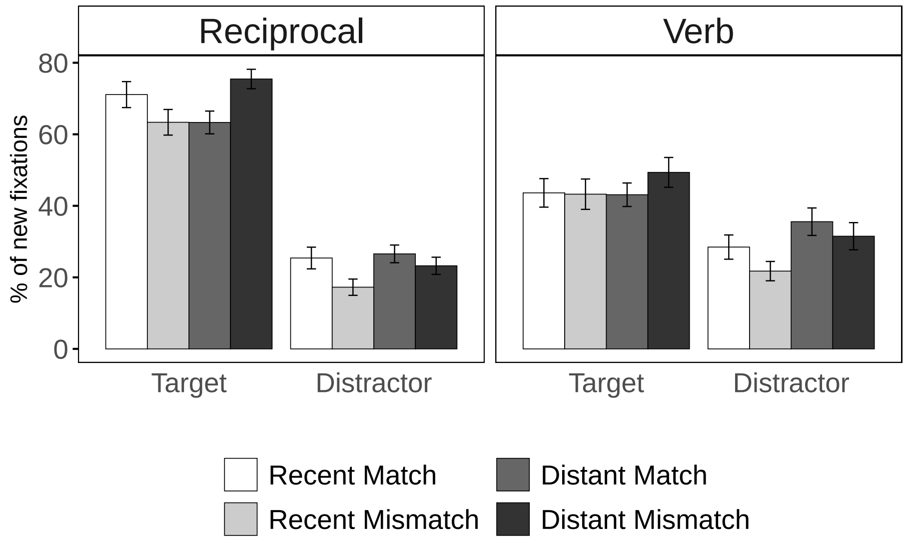


Figure 5: Mean percentages of new fixations towards the target and distractor out of all new fixations within the reciprocal and verb windows in Experiment 1. Error bars show SEs over by-participant means.

the target or the distractor, averaged across the four experimental conditions. This does not test any hypothesis of interest, so although we report the intercepts for completeness, we do not directly interpret them. For new fixations towards the target, the analyses for the reciprocal window showed a significant interaction between RECENCY and MATCH. The nested model showed that in conditions with a linearly distant target, there were fewer new fixations towards the target in the presence of a plural as opposed to a singular distractor, but the distractor number did not affect the fixations when the target was linearly recent. In the verb window, the proportion of a new fixation was not modulated by any of the factors.

For new fixations towards the distractor, the analyses for the reciprocal window showed a non-significant trend for more fixations towards a plural distractor. The nested model shows that this was driven by the recency of the target: In conditions with a linearly recent target, there were more fixations towards a linearly distant, plural distractor than a singular one, but there was no such difference when the target was distant. In the verb window, there were fewer fixations towards the distractor when it was linearly distant (i.e., when the target was linearly more recent).

*3.6. Discussion*

Experiment 1 investigated the role of the c-command relation in processing the local anaphor *birbirleri* in Turkish by employing stimuli that included c-commanding target subjects and non-c-commanding distractors in the same clause[5], marked with the same case marking. Despite neutralizing these potential cues to the target, the results show that c-commanding subjects were immediately distinguished from distractors immediately at the reciprocal anaphor, as revealed by increased gaze proportions towards the target over the distractor. This effect appears regardless of whether the target linearly precedes or follows the distractor, with a further advantage for linearly recent targets. Together, these findings suggest that antecedent retrieval is guided by fine-grained hierarchical information about NP relations within a sentence, as captured in the c-command relationship.

We also contrasted singular and plural distractors to reveal whether morphological number match between the anaphor and the distractor would modulate the retrieval process (e.g., Badecker and Straub, 2002). We see only limited evidence that number-matching distractors are reactivated during antecedent retrieval, as evidenced by the overall increased availability of target subjects over distractors. Moreover, the evidence for feature-based interference is not consistent, with effects varying depending on, but not consistently shaped by, the linear order of target and distractor. Specifically, we see a decrease in total gaze proportions and proportions of a new fixation towards target subjects in the presence of a plural, linearly more recent distractor. Conversely, we observe an increase in the proportions of new fixations towards the plural distractor when it is linearly more distant than the target[6]. Given the mixed nature of these results, we sought to establish whether they would be replicated in Experiment 2 before interpreting them further.

Although we have highlighted the central importance of c-command in distinguishing the target from the distractor in our stimuli, this design alone does not

[5]A reviewer points out that the structure in (5-a) could potentially be parsed as a doubly-embedded sentence with three consecutive subjects at the anaphor, if a silent *pro* is assumed as the possessor of the target noun *cameramen*. Under this parse the distractor *directors* would not be a clause-mate antecedent to the target. While we cannot rule out this possibility, we think this parse is unlikely to be (strongly) committed to during online comprehension due to its high complexity (e.g., Inoue and Fodor, 1995). Moreover, (5-b) unambiguously places the distractor in the same clause as the target, supporting our conclusion regarding clause-mateness in Experiment 1.

[6]As noted by a reviewer, an alternative explanation for the observed recency effects in the Recent Target conditions is that they may reflect the cost of reanalyzing the structural role of the distractor (see footnote 5). Although the critical anaphor region is two words after the target, the relevant processing cost may potentially spill over, affecting overall fixations at the anaphor.

unambiguously support the conclusion that the c-command relation is the relevant relationship that guides early antecedent retrieval processes. One important caveat is that in Experiment 1's stimuli, the target antecedent was always the subject of the clause that included the anaphor. Thus, an alternative explanation could be that the target is retrieved through a combination of two item-level features, subject (e.g., genitive case) and clause features (e.g., local), without a need for exploiting hierarchical relations like c-command (although as Franck and Wagers (2020) note, 'subject of the current clause' is arguably a relational notion). This suggests that further work is needed to unambiguously establish c-command as a key factor in antecedent retrieval. We investigate this by examining the availability of c-commanding antecedents in object position in Experiment 2. Moreover, we note that the stimuli in Experiment 1 were not pre-tested for the plausibility of the subject vs. distractor NPs serving as an object for the verb. This is a potential confound, as a semantic plausibility disadvantage for the distractor could have influenced the results in favor of the subject. We address this concern in Experiments 2 and 3 by using the same NPs as both a potential target and a distractor.

## 4. Experiment 2: C-command and subjecthood

Experiment 2 investigates whether the immediate retrieval of c-commanding antecedents in subject position observed in Experiment 1 extends to c-commanding antecedents in other syntactic positions; specifically those in indirect object position. In Experiment 1, and indeed, in all previous studies that we reviewed in the Introduction, the target antecedent appeared as the subject of the clause containing the anaphor. Thus it is possible that a composite cue like *subject-of-the-current-clause* underwrites antecedent retrieval for anaphors (Arnett and Wagers, 2017; Kush and Phillips, 2014; Schoknecht et al., 2025; Xiang et al., 2009). Under this scenario, the robust availability of target subjects immediately within the reciprocal region in Experiment 1 could reflect the use of subject and clause information rather than sensitivity to hierarchical relations between NPs.

However, local anaphors can be bound by arguments other than subjects in many languages including Turkish (Göksel and Kerslake, 2005; Kornfilt, 1997; Legate et al., 2020). A stronger test of the claim that c-command is used online to retrieve antecedents for anaphors could leverage this fact to test whether comprehenders also distinguish non-subject targets from distractors on the basis of c-command, rather than by grammatical role/subjecthood. To this end, in Experiment 2 we test c-commanding targets in indirect object position, thereby disentangling the effect of argument role in the local clause from c-command. Like in Experiment 1, c-commanding targets and distractors are in the same clause to eliminate clausal information as a distinguishing cue. Additionally, both the indirect object and the distractor were marked with dative case to neutralize case information.

We predicted that if the c-command relation between NPs guides antecedent retrieval, then we would observe increased looks to c-commanding subjects as well as c-commanding indirect objects when compared to non-c-commanding distractors.

*4.1. Materials*

Experiment 2 again tests *birbirleri* as the direct object of an embedded clause, as in (8). All sentences have a c-commanding, clause-mate antecedent in the subject position and an additional dative-marked noun in the same clause. There are five conditions in a within-subject design. The conditions in (8-a)-(8-b) have a 2x2 design with a plural subject, crossing the C-COMMAND relation of the dative-marked noun with respect to the anaphor, i.e., c-commanding indirect object (IO) vs. non-c-commanding distractor, and number MATCH of this referent with the anaphor, i.e., singular vs. plural. In hierarchical terms, the phrases of the embedded subject and IO are a sister to the node containing the anaphor; hence, they c-command the anaphor. However, the phrase of the distractor is embedded in a PP and is not a sister to the same node; hence, it does not c-command the anaphor. The fifth condition has a singular subject and a plural IO, as shown in (8-c), and was added as a control condition to investigate the overall availability of IO in the absence of a plural subject. In (8)[7] we mark all c-commanding subjects and IOs as available antecedents through coindexation as they can occasionally bind the reciprocal regardless of their number (mis)match (see below).

(8) Example item from Experiment 2 (with anaphor in bold, target(s) underlined and distractor in italic)

a. Plural Subject, IO Match / Mismatch

| | | | | |
|---|---|---|---|---|
| Ayşe$_i$ | [kameraman-lar-ın$_j$ | yönetmen(-ler)-e$_k$ | açıkça | **birbirlerin-i**$_{*i/j/k}$ |
| Ayşe | cameraman-PL-GEN | director-PL-DAT | openly | each.other-ACC |

| | |
|---|---|
| sor-duğ-un]-u | anla-dı. |
| ask-NMLZ-3SG.POSS-ACC | notice-PST |

[7]The Distractor conditions as in (8b) (and those in Experiment 3) can alternatively be labeled as *Adverb* or *Adjunct* to better describe their structural role in the sentence, particularly compared to that of the IO. In our presentation of the stimuli we used the term *Distractor* as it is commonly used in anaphor processing literature.

‘Ayşe$_i$ noticed that the cameramen$_j$ openly asked the director(s)$_k$ about each other$_{*i/j/k}$.’

b. Plural Subject, Distractor Match / Mismatch

| | | | | |
|---|---|---|---|---|
| Ayşe$_i$ | [kameraman-lar-ın$_j$ | *yönetmen(-ler)-e*$_k$ | göre | açıkça |
| Ayşe | cameraman-PL-GEN | director-PL-DAT | according.to | openly |
| **birbirlerin-i**$_{*i/j/*k}$ | sor-duğ-un]-u | | anla-dı. | |
| each.other-ACC | ask-NMLZ-3SG.POSS-ACC | | notice-PST | |

‘Ayşe$_i$ noticed that the cameramen$_j$, according to the director(s)$_k$, openly asked about each other$_{*i/j/*k}$.’

c. Singular Subject, IO Match

| | | | | |
|---|---|---|---|---|
| Ayşe$_i$ | [kameraman-ın$_j$ | yönetmen-ler-e$_k$ | açıkça | **birbirlerin-i**$_{*i/j/k}$ |
| Ayşe | cameraman-GEN | director-PL-DAT | openly | each.other-ACC |
| sor-duğ-un]-u | anla-dı. | | | |
| ask-NMLZ-3SG.POSS-ACC | notice-PST | | | |

‘Ayşe$_i$ noticed that the cameraman$_j$ openly asked the directors$_k$ about each other$_{*i/j/k}$.’

Both the IO and the distractor follow the embedded subject, are clause-mates to the anaphor, and have dative case *-(y)A*. The dative noun forms an adjunct with the postposition, e.g., *göre* ‘according to’, in the Distractor conditions, and it functions as an argument of the embedded verb in the IO conditions. The interpretation of the anaphor in the IO condition is ambiguous because there are two available antecedents in c-commanding positions: the embedded subject and the IO. There is only one available antecedent in the Distractor conditions: the embedded subject. The embedded verb, e.g., *sor-* ‘to ask’, is optionally ditransitive, allowing two object arguments in the IO conditions and only a direct object (DO) argument in the Distractor conditions.

Thirty sets of the five conditions in (8) were created as experimental stimuli and distributed across five lists in a Latin Square design. The experimental sentences were interspersed with 72 filler items that had a similar format to those in Experiment 1: The anaphor was replaced with the unmentioned singular in 24 sentences, with the unmentioned plural in another 24 sentences, and with the distractor in 8 sentences; The remaining filler items included 10 sentences where the verb took a dative-marked object which was intended to be the answer to the question and 6 sentences with a passivized verb where the answer to the question was intended to be the subject.

Before the visual world study, an offline norming task confirmed that the indirect object c-commanded the anaphor with the verbs in our stimuli (Öztürk, 2005) (see Appendix B.2). Moreover, two offline antecedent selection tasks verified that *birbirleri* can be bound by both subject and IO positions, but not the distractors (for details, see Appendix B.3 and Appendix B.4). We found that plural IOs are more frequently selected as antecedents than singular IOs, confirming their availability as antecedents. However, the overall selection rate of IOs is smaller than that of subjects (i.e., 35% vs. 91% & 20% vs. 85%), suggesting a subject bias in interpreting the reciprocal. Singular IOs were occasionally selected as antecedents, and singular subjects were chosen more often than plural IOs in the control condition. This suggests that c-command and subjecthood may play a more dominant role in antecedent selection than number alone. While this deserves further investigation, we proceeded with these stimuli, for which c-command does appear to be a crucial constraint in antecedent selection for *birbirleri*.

### *4.2. Participants*

Fifty-four native speakers of Turkish ($M_{age} = 22.7$) from the same pool participated in the study. All participants had normal or corrected-to-normal vision. They received 300 Turkish liras (∼$10 at the time of the study) for compensation. An inspection of participants' accuracy in the filler items showed that two participants' accuracy was below 70%; so they were removed from analyses, leaving 52 participants for analyses.

### *4.3. Procedure*

The trial presentation and the format of the lead-in sentences and questions were the same as in Experiment 1. As in Experiment 1, the question targeted the noun bearing the thematic role of the anaphor which may appear in different grammatical positions such as subject or IO. An example lead-in sentence and question for the experimental stimuli in (8) are in (9) below.

(9) a. Lead-in

| | | | |
|---|---|---|---|
| Ayşe | meraklı | soru-lar | duy-muş. |
| Ayşe | curious | question-PL | hear-PST |

'Ayşe heard curious questions.'

b. Question

| | | | |
|---|---|---|---|
| Ayşe | kim-in | sor-ul-duğ-un-u | anla-dı? |
| Ayşe | who-GEN | ask-PASS-NMLZ-3SG.POSS-ACC | notice-PST |

'Who were the people that Ayşe noticed were asked about?'

*4.4. Data analyses*

The data in Experiment 2 were preprocessed and analyzed as in Experiment 1. The critical comparisons in the analyses differ from those in Experiment 1 in the following way. We performed separate statistical analyses for the 2x2 design, i.e., conditions with a plural subject as in (8-a)-(8-b), and the control condition, i.e., the Singular Subject with IO Match condition as in (8-c). For the 2x2 design, fixed effects were C-COMMAND and MATCH of the dative-marked noun, with random effects for participants and items. Both fixed effects were sum-contrast coded and included in a crossed and nested model that were fit to the data. The crossed model included both fixed effects and their interaction, and the nested model had a main effect of C-COMMAND and separate MATCH contrasts within the IO and Distractor conditions. Table 4 provides contrast coding for the 2x2 design in both models. The rest of the model specifications are the same as in Experiment 1. The control condition was analyzed with a *t*-test comparing the looks to the singular subject vs. the plural IO. The average durations of the reciprocal and verb in Experiment 2 are 632 and 630 ms, respectively.

| | Crossed Model | | | Nested Model | | |
|---|---|---|---|---|---|---|
| | C-command | Match | C-command:Match | C-command | IO: Match | Distractor: Match |
| IO Match | 0.5 | 0.5 | 0.25 | 0.5 | 0.5 | 0 |
| IO Mismatch | 0.5 | -0.5 | -0.25 | 0.5 | -0.5 | 0 |
| Distractor Match | -0.5 | 0.5 | -0.25 | -0.5 | 0 | 0.5 |
| Distractor Mismatch | -0.5 | -0.5 | 0.25 | -0.5 | 0 | -0.5 |

Table 4: Contrast coding coefficients used for the crossed and nested statistical models for the 2x2 design in Experiment 2. Experimental conditions are given in rows, model contrasts given in columns.

*4.5. Results*

Participants' mean accuracy across filler items was 93.3%. Trials that had a track loss of 50% or more were removed, which resulted in the loss of 0.008% of experimental items.

*4.5.1. Picture selection data*

The participants' selections of pictures in the experimental items are in Table 5. The embedded subject was chosen around 81.2% of the time, the IO 25.5% of the time and the distractor 5.9% of the time.

A mixed-effects logistic regression model fit to the selection data for the dative-marked noun in the 2x2 design showed a main effect of C-COMMAND, with more choices of IO than distractor ($z$'s $\geq 3.57$); there were no other effects ($z$'s $\leq 1.66$). A binomial test on the proportion of subject and IO responses in the control condition showed that subject and IO responses were not significantly different ($p = 0.91$, 95% CI: 0.45 - 0.56%).

| | Subject % | IO/ Distractor % | Unmentioned Plural % | Singular % |
|---|---|---|---|---|
| Plural Subject, IO Match | 82.1 (2.4) | 16.3 (2.3) | 0.6 (0.4) | 1 (5.4) |
| Plural Subject, IO Mismatch | 88.3 (2.3) | 11.3 (2.3) | 0 | 0.3 (0.3) |
| Plural Subject, Distractor Match | 91.8 (2) | 6.9 (1.7) | 0.3 (0.3) | 1 (0.5) |
| Plural Subject, Distractor Mismatch | 93.8 (1.8) | 4.8 (1.6) | 0.4 (0.4) | 1 (0.5) |
| Singular Subject, IO Match | 49.7 (3.9) | 48.7 (4.1) | 0.6 (0.4) | 1% (0.5) |

Table 5: Mean percentages (and SEs over by-participant means) for picture selections in Experiment 2.

*4.5.2. Gaze proportion analyses*

Figure 6 plots the mean raw gaze proportions towards the four interest areas in each condition, spanning a period of 1500 ms with time slices of 5 ms (for gaze proportions for the entire sentence, see Appendix F).

As in Experiment 1, the analyses were conducted on the corrected log subject vs. IO/distractor ratios that were calculated by subtracting the corrected gaze proportion of IO or distractor from that of the subject on the log scale, for the reciprocal and verb windows separately. Figure 7 shows this measure for the two time windows. Corresponding analyses for the 2x2 design are in Table 6.

For the reciprocal window, there was a positive intercept in both crossed and nested models, suggesting that there were overall more looks towards the subject than IO/distractor. This subject bias was around 2.5 times stronger in the verb window than in the reciprocal. In both time windows, the dative noun received more looks when it was a c-commanding IO compared to when it was a distractor. There were no effects of number match or an interaction.

For the control condition, a $t$-test showed that the singular subject vs. plural IO ratio was not reliably different from 0, suggesting no bias towards the subject or IO in the reciprocal ($t$ = -1.65) or verb window ($t$ = -0.39).

*4.5.3. New fixation analyses*

Figure 8 shows the proportions of new fixations towards the subject and IO/distractor out of all new fixations within the reciprocal and verb windows.

The corresponding analyses on the proportions of new fixations are in Table 6. The results showed that the probability of a new fixation towards the subject was not modulated by any of the experimental factors. For new fixations towards the

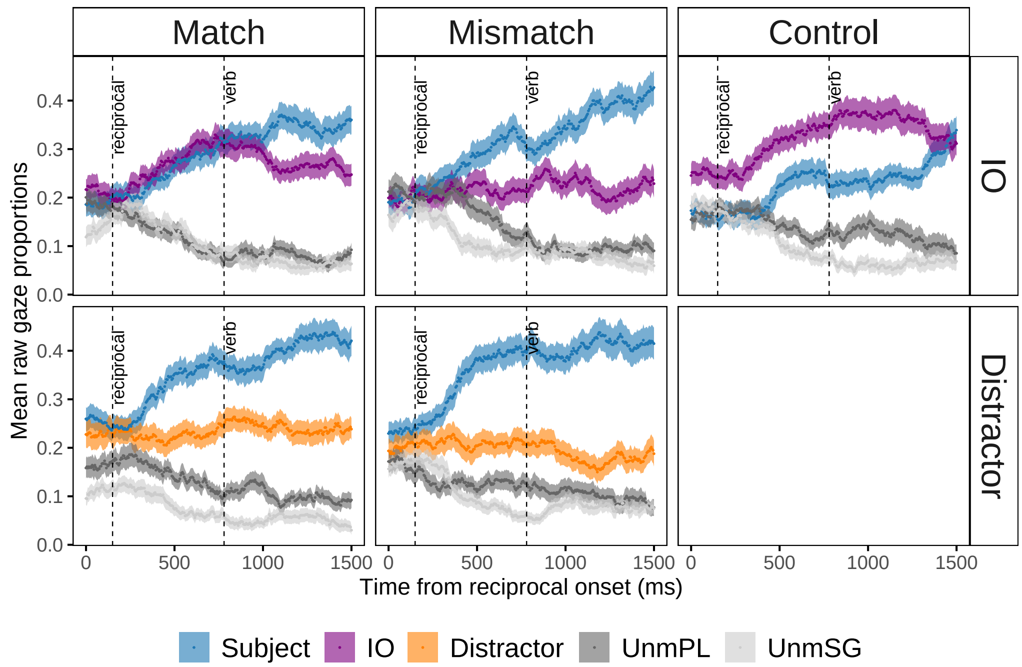


Figure 6: Mean raw gaze proportions towards the four interest areas in Experiment 2. Error bands show SEs over by-participant means. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Proportions are time-locked to the onset of the critical anaphor and span 1500 ms. Dashed lines mark the 150-ms shifted onsets of the reciprocal and verb windows.

IO/distractor, the analyses on the reciprocal window showed that there were more fixations when the dative noun was plural, and there was a significant interaction between C-COMMAND and MATCH. The nested model showed that this interaction was driven by the c-commanding status of the dative noun: There were more fixations towards the plural than the singular, c-commanding IO; but there was no difference between singular and plural distractors.

For the control condition, binomial tests compared the proportions of new fixations towards the subject vs. IO. Results showed that there were more fixations towards IO than the subject in the reciprocal window ($p < 0.01$, 95% CI: 0.52 - 0.67%); but fixations towards the subject and IO were not significantly different in the verb window ($p = 0.64$, 95% CI: 0.44 - 0.60%).

### *4.6. Discussion*

Experiment 2 tested whether the immediate availability of c-commanding subjects at retrieval extends to c-commanding indirect objects (IOs), a key prediction of

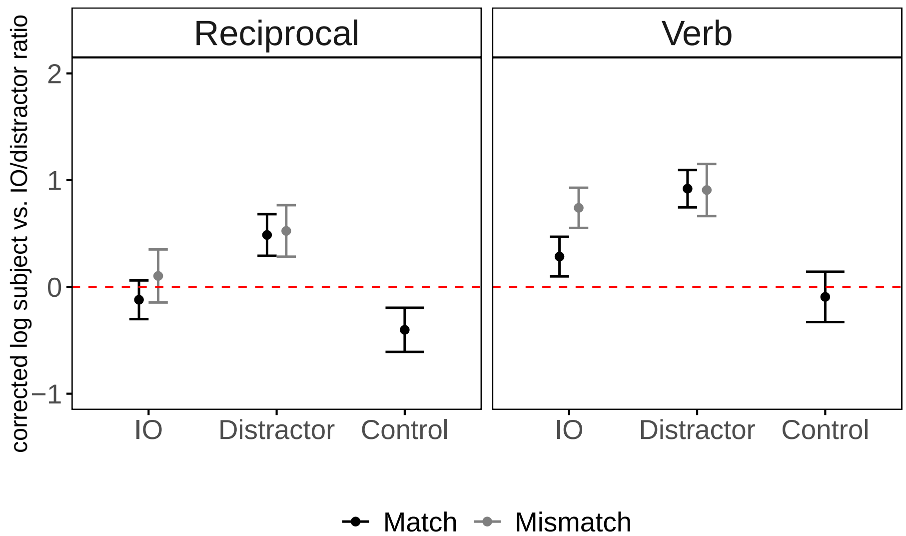


Figure 7: Mean of corrected embedded subject vs. IO/distractor ratios on the log scale for the reciprocal and verb windows in Experiment 2. Error bars show SEs over by-participant means. Red dashed lines at 0 indicate equal looks to the subject and IO/distractor; values above 0 reflect greater looks to the subject and those below 0 greater looks to the IO/distractor.

the hypothesis that hierarchical/relational processing supports antecedent retrieval associated with anaphors. We observe that IOs were rapidly differentiated from non-c-commanding distractors at the reciprocal, despite having neutralized case and clausehood cues. This provides evidence that memory retrieval targets antecedents that are in a hierarchically available position with respect to the anaphor, rather than simply targeting the subject of the current clause.

Interestingly, our results suggest that c-commanding singular antecedents, particularly subjects, can occasionally bind the reciprocal despite a number mismatch. In offline antecedent selection tasks, plural IOs were selected more frequently than singular IOs, confirming the reciprocal's sensitivity to number. This pattern was partly supported by the visual world data: While gaze proportions do not differ between singular and plural IOs, participants made more new fixations to plural IOs in the reciprocal window. However, in the control condition comparing singular subjects and plural IOs, singular subjects were selected more often than plural IOs in antecedent selection tasks. The online data show no bias for either in gaze proportions, but more new fixations towards the plural IO in the reciprocal window. Overall, these findings suggest a strong preference for c-commanding antecedents, especially for those in subject position, a preference that may potentially override number cues.

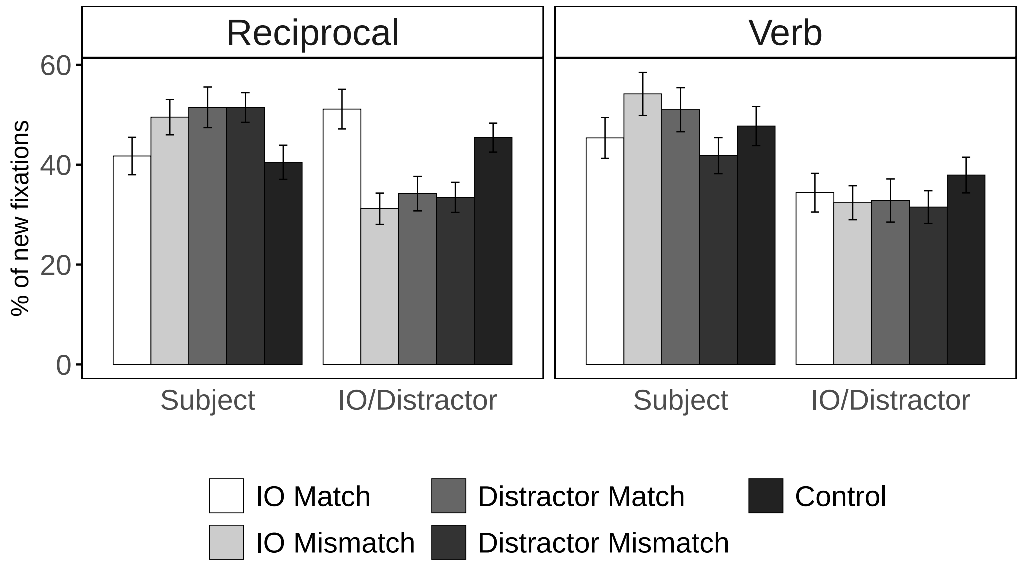


Figure 8: Mean percentages of new fixations towards the embedded subject and IO/distractor out of all new fixations within the reciprocal and verb windows in Experiment 2. Error bars show SEs over by-participant means.

Before strongly interpreting these findings, however, we test whether they would replicate in a higher-powered study (e.g. Nicenboim et al., 2018).

## 5. Experiment 3: High-powered, pre-registered replication

Experiment 3 aimed to determine whether the previously observed retrieval advantage for c-commanding subjects and IOs, compared to non-c-commanding distractors, would replicate in a higher-powered study.

To this end, we selected three conditions from Experiments 1-2 that compared the availability of c-commanding plural subjects followed by either (i) c-commanding plural IOs or (ii) non-c-commanding plural distractors. To maximize statistical power, only conditions involving plural targets and distractors were included. We compared genitive- or dative-marked distractors from Experiments 1 and 2, respectively, to examine whether the distractors' differing case properties/structural role would impact antecedent retrieval beyond their syntactically subordinated status. Additionally, for the dative-marked distractors, we introduced two new postpositions (see Appendix B.3 and Appendix B.4).

The experiment was pre-registered on OSF at the following link: https://osf.io/s9f85.

*5.1. Materials*

Similar to Experiments 1-2, Experiment 3 employed Turkish sentences in which *birbirleri* was the direct object of an embedded clause. All sentences have a c-commanding, clause-mate subject and an additional, second NP in the embedded clause, as in (10). There are three conditions, manipulating the C-COMMAND relation of this additional referent, i.e., c-commanding IO vs. non-c-commanding distractor, and CASE marking on the distractor, i.e., dative vs. genitive. The conditions in (10-a)-(10-b) have the same structure as the Plural Subject, IO Match and Plural Subject, Distractor Match conditions from Experiment 2, respectively (see (8-a)-(8-b) above), and the one in (10-c) has the same structure as the Distant Target, Distractor Match condition from Experiment 1 (see (5-b) above). The conditions are relabeled here for consistency within the experiment.

(10) Example item from Experiment 3 (with anaphor in bold, target(s) underlined and distractor in italic)

a. IO

Ayşe$_i$ [aşçı-lar-ın$_j$ garson-lar-a$_k$ sebepsizce **birbirlerin-i**$_{*i/j/k}$
Ayşe chef-PL-GEN waiter-PL-DAT without.reason each.other-ACC
kötüle-dik-lerin]-i bil-iyor.
badmouth-NMLZ-3PL.POSS-ACC know-PROG

'Ayşe$_i$ knows that the chefs$_j$ badmouthed each other$_{*i/j/k}$ to the waiters$_k$ for no reason.'

b. Distractor, Dative Case

Ayşe$_i$ [aşçı-lar-ın$_j$ *garson-lar-a*$_k$ rağmen sebepsizce
Ayşe chef-PL-GEN waiter-PL-DAT despite without.reason
**birbirlerin-i**$_{*i/j/*k}$ kötüle-dik-lerin]-i bil-iyor.
each.other-ACC badmouth-NMLZ-3PL.POSS-ACC know-PROG

'Ayşe$_i$ knows that the chefs$_j$, despite the waiters$_k$, badmouthed each other$_{*i/j/*k}$ for no reason.'

c. Distractor, Genitive Case

Ayşe$_i$ [aşçı-lar-ın$_j$ *garson-lar-ın*$_k$ etki-si-yle
Ayşe chef-PL-GEN waiter-PL-GEN influence-3SG.POSS-COM
sebepsizce **birbirlerin-i**$_{*i/j/*k}$ kötüle-dik-lerin]-i
without.reason each.other-ACC badmouth-NMLZ-3PL.POSS-ACC
bil-iyor.
know-PROG

'Ayşe$_i$ knows that the chefs$_j$, with the waiters'$_k$ influence, badmouthed each other$_{*i/j/*k}$ for no reason.'

Both the IO and distractor follow the embedded subject and are clause-mates to the anaphor. The IO is marked with the dative case *-(y)A*, and is an argument of a ditransitive verb, e.g., *kötüle- 'to badmouth'*. The distractor is either marked with the dative case *-(y)A* and followed by a postposition, e.g., *rağmen* 'despite', forming an adverbial clause, or with the genitive case *-(n)In* and embedded in a genitive-possessive construction as a possessor noun. Whereas the c-commanding subject and the distractor have different case markings in (10-b), they are marked with the same genitive case in (10-c). For (10-b), two postpositions that lexically required dative case on their complement were used in the stimuli; *karşın* 'despite' and *rağmen* 'despite'. As in Experiment 2, the verb, e.g., *kötüle-* 'to badmouth', is optionally ditransitive, allowing it to have two object arguments in the IO conditions and only a DO argument in the Distractor conditions. The same verbs, normed for the hierarchical relationship between direct and indirect objects in Experiment 2, are used across conditions. The interpretation of the anaphor in the IO condition is ambiguous because there are two available antecedents in c-commanding positions: the embedded subject and the IO.

Twenty-four sets of the three conditions were created and distributed across three lists in a Latin Square design. In addition, there were 72 filler items that had a similar format to those in Experiments 1 and 2: The anaphor was replaced with the unmentioned singular in 24 sentences, with the unmentioned plural in another 24 sentences, and with the distractor in 16 sentences. In the remaining 8 sentences, the verb took a dative-marked object which was intended to be the answer to the question.

Before the visual world study, an offline antecedent selection task verified that *birbirleri* can be bound by the c-commanding subject or IO in this set of stimuli, with each selected 95% and 46% of the time, respectively (for details, see Appendix B.5). The overall low selection rate of distractors (i.e., 16%) shows that they are unavailable as antecedents to bind the reciprocal.

*5.2. Participants and power analysis*

We collected data from 102 native speakers of Turkish ($M_{age}$ = 22.1) from the same pool for Experiment 3 (see Appendix G for power analysis justifying sample size). All participants had normal or corrected-to-normal vision. Each participant received 300 Turkish liras (∼$10 at the time of the study) for compensation. An inspection of participants' accuracy in the filler items shows that one participant's accuracy was below 70%, so they were removed from analyses. Data from one participant were excluded due to major track loss in more than 70% of the experimental sentences, leaving 100 participants for analysis.

### 5.3. Procedure

The trial presentation and the format of the lead-in sentences and questions preceding the experimental items are the same as in previous experiments.

### 5.4. Data analyses

The data in Experiment 3 were pre-processed and analyzed for total gaze proportions and proportions of a new fixation following the same steps in Experiments 1-2. The critical comparisons in the analyses differ from those in previous experiments in the following way. There is only one experimental factor, second noun (NP2). This was coded using Helmert contrasts, which compared the effects of the C-COMMAND relation of the second noun, i.e., IO vs. the two Distractor conditions, and CASE marking on the distractor, i.e., dative vs. genitive. Contrast coding is in Table 7. We fit a model with these contrasts and random intercepts for participants and items, and random slopes when models converged. We again analyzed gaze proportions and new fixations. The average durations of the reciprocal and verb in Experiment 3 are 659 and 752 ms, respectively.

| | C-command | Case |
|---|---|---|
| IO, Dative | 0.66 | 0 |
| Distractor, Dative | -0.33 | 0.5 |
| Distractor, Genitive | -0.33 | -0.5 |

Table 7: Helmert contrast coding coefficients used for the mixed effects models in Experiment 3. Experimental conditions are given in rows, model contrasts given in columns.

### 5.5. Results

Participants' overall accuracy in the filler items was 95.5%. Trials that had a track loss of 50% or more were removed, resulting in the loss of 0.009% of the data for experimental sentences.

#### 5.5.1. Picture selection data

The participants' selections of pictures in the experimental items are in Table 8. Overall, the subject was chosen around 87.5% of the time, the IO 20% of the time and the distractor around 8% of the time.

| | Subject % | IO/ Distractor % | Unmentioned | |
|---|---|---|---|---|
| | | | Plural % | Singular % |
| IO, Dative | 79.2 (2.2) | 20.1 (2.2) | 0.5 (0.3) | 0.1 (0.1) |
| Distractor, Dative | 92.9 (1.1) | 6.6 (1.1) | 0.4 (0.2) | 0.1 (0.1) |
| Distractor, Genitive | 90.4 (1.2) | 9.2 (1.2) | 0.4 (0.2) | 0 |

Table 8: Mean percentages (and SEs over by-participant means) for picture selections in Experiment 3.

A mixed-effects logistic regression model fit to the data for the NP2 choices revealed a main effect of C-COMMAND, with more choices of IO than distractor ($z = 4.51$). A comparison within the two Distractor conditions showed that the genitive-marked distractor was chosen more often than the dative-marked distractor ($z = 2.06$). The increased choices of the genitive-marked distractors may be due to their case similarity to c-commanding subjects (also see Bakay, 2020; Lago et al., 2019; Slioussar et al., 2022; Türk and Logačev, 2024). Regardless, the overall choices of the distractors were quite low.

*5.5.2. Gaze proportion analyses*

Figure 9 plots the mean raw gaze proportions towards the four interest areas in each condition, spanning a period of 1500 ms with time slices of 5 ms (see Appendix H for gaze proportions for the entire sentence).

The dependent measure was the corrected log subject vs. IO/distractor ratios calculated as in Experiment 2, as illustrated in Figure 10 below. Corresponding analyses are in Table 9.

In both reciprocal and verb windows, there was a positive intercept, suggesting overall more looks towards the subject over IO/distractor. The subject bias was stronger in the verb window. In both time windows, the second NP received more looks when it was a c-commanding IO compared to when it was a distractor. There was no difference between the two Distractor conditions.

*5.5.3. New fixation analyses*

Figure 11 shows the proportions of new fixations towards the subject and IO/distractor out of all new fixations in the reciprocal and verb windows. Corresponding analyses are in Table 9.

For the analyses on new fixations towards the subject, there were fewer fixations towards the subject when the second noun was an IO compared to when it was a distractor. For the analyses on new fixations towards the second noun, there were more new fixations on an IO than a distractor in the reciprocal window.

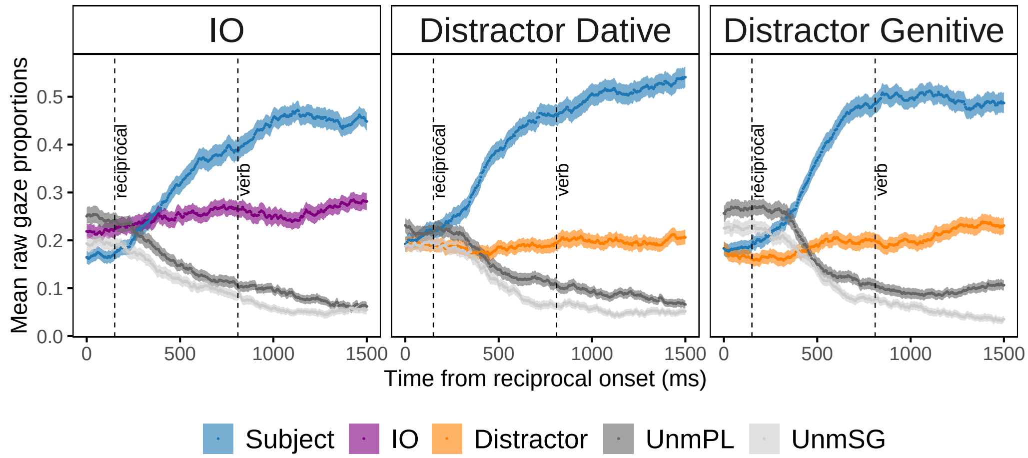


Figure 9: Mean raw gaze proportions towards the four interest areas in Experiment 3. Error bands show SEs over by-participant means. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Proportions are time-locked to the onset of the critical anaphor and span 1500 ms. Dashed lines mark the 150-ms shifted onsets of the reciprocal and verb window.

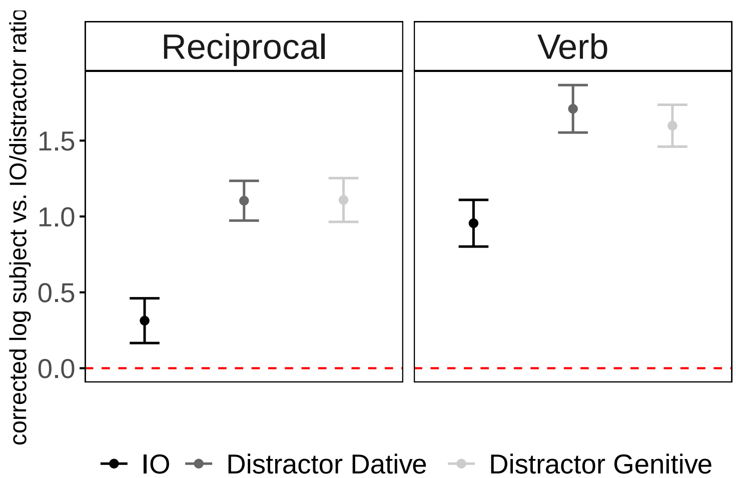


Figure 10: Mean of corrected embedded subject vs. IO/distractor ratios on the log scale for the reciprocal and verb windows in Experiment 3. Error bars show SEs over by-participant means. Red dashed lines at 0 indicate equal looks to the subject and IO/distractor; values above 0 reflect greater looks to the subject.

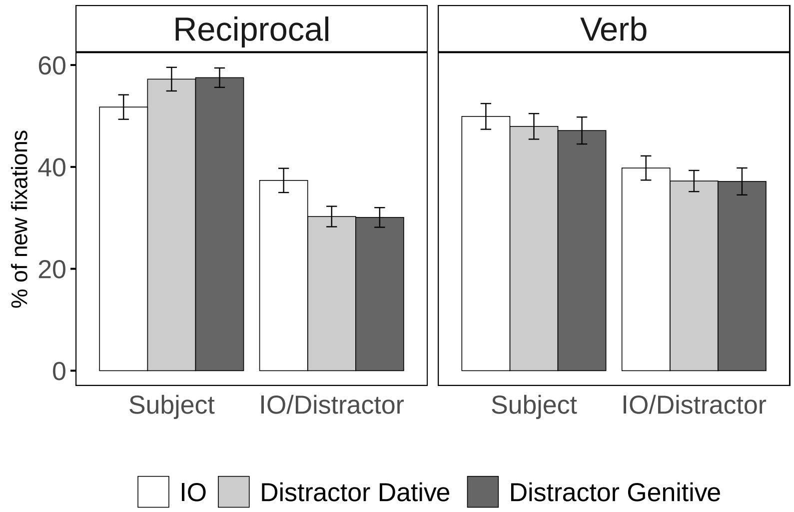


Figure 11: Mean percentages of new fixations towards the embedded subject and IO/distractor out of all new fixations within the reciprocal and verb windows in Experiment 3. Error bars show SEs over by-participant means

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*5.6. Discussion*

Experiment 3 was a high-powered, pre-registered replication of the key contrasts from Experiments 1-2, testing whether the immediate availability of c-commanding antecedents would replicate with a larger sample size. As in Experiments 1-2, the target antecedents and distractors are in the same clause, preventing the use of clause-level cues to distinguish them. Differently from Experiments 1-2, Experiment 3 directly compared two different distractors which differed in terms of their case similarity with the target and their structural role in the sentence.

The results of Experiment 3 replicate previous findings that c-commanding targets in subject and indirect object positions are differentiated from non-c-commanding distractors immediately within the reciprocal window by receiving more looks and attracting more new fixations. Looks to the two distractors were comparable regardless of their case/structural roles. These findings suggest a robust and immediate role of hierarchical relations between NPs in antecedent retrieval above and beyond item-level structural cues.

## 6. General Discussion

In three experiments we tested whether the initial stages of antecedent retrieval are guided by hierarchical relations between NPs in a sentence by investigating the real-time resolution of the local, preverbal reciprocal anaphor *birbirleri* in Turkish. We examined the time-course of reactivation of different potential antecedents for the critical anaphor. In Experiment 1, we saw overall more looks to c-commanding subjects than case-matching, clause-mate distractors immediately at the reciprocal anaphor region. This effect was greater when the subject was linearly closer to the anaphor than the distractor. We also saw limited, inconsistent evidence for feature-based interference from plural distractors. In Experiment 2, we again saw overall increased looks to c-commanding subjects over non-c-commanding distractors, and further saw increased looks to c-commanding indirect objects relative to case-matching, clause-mate but non-c-commanding distractors. Crucially, analyses on new fixations revealed that more new fixations were launched towards the plural indirect object than the singular one. The number feature of the distractor did not modulate new fixations. Experiment 3 was a pre-registered, high-powered replication of the key contrasts from Experiments 1-2; In Experiment 3 we again saw that c-commanding subjects and indirect objects were immediately reactivated at the anaphor to a larger degree than non-c-commanding distractors.

Overall, these findings provide novel evidence that hierarchical relations among NPs in a clause, as operationalized in the c-command constraint, are an important determinant of the availability of antecedents for local anaphors in early stages of memory retrieval. This extends previous observations by Cunnings et al. (2015); Franck and Wagers (2020); Kush et al. (2015); Moulton and Han (2018) with a novel methodology (visual world), and shows that the c-command advantage at retrieval extends to within clause hierarchical relations between NPs. Importantly, these results establish the importance of relational hierarchical structure above and beyond inherent syntactic features of an item in memory, such as clause, case and subjecthood, and domain-general memory factors such as linear order/recency. Finally, these findings also extend previous observations by showing that non-subject NPs (here, indirect objects) that are in a hierarchically prominent position have a similar advantage at retrieval to c-commanding subjects.

Overall, then, our results are consistent with a tight relationship between the hierarchical relational constraints on a given dependency (here, reciprocal anaphors), and the real-time reactivation of potential antecedents at the point of processing a dependent anaphor (Kush, 2013). Hierarchical, relational information, such as the c-command constraint, guides antecedent retrieval processes from the earliest stages. This is inconsistent with the view that such constraints reflect a second-order, inferred relationship between items in memory, which would erroneously predict that they should become available relatively late in processing (McElree and Dosher, 1993).

While these results suggest that hierarchical syntactic relations are an important component of online retrieval, there are a range of different ways this could be implemented. We have highlighted c-command as the relevant notion of hierarchy, but an alternative interpretation of our findings might instead focus on coargumenthood. Since the c-commanding antecedents in our stimuli also functioned as coarguments of the anaphor (i.e., they are all in the same verbal domain), our results could reflect a retrieval mechanism that primarily draws on argument structure information (Wagers, 2008). Indeed, some linguistic theories identify this as the critical hierarchical notion central to anaphor binding (Pollard and Sag, 1994; Reinhart and Reuland, 1993). Argument structure could plausibly guide retrieval preverbally, since coarguments can be identified via case marking or the emerging syntactic parse.

While the present results do not determine whether the hierarchical constraints on retrieval are determined in terms of syntactic geometry (as in c-command) or verbal coargumenthood, we think the former is more plausible in light of a broader body of psycholinguistic evidence. As discussed in detail in Section 1.1, previous studies by Cunnings et al. (2015); Kush et al. (2015); Moulton and Han (2018) provided evidence for an immediate role of c-command in processing bound variable pronouns that cannot be reduced to coargumenthood. C-command thus offers a unifying explanation for the previous results and ours. We leave further experimental tests to directly disentangle c-command from co-argumenthood to future work.

### *6.1. Hierarchical information in memory retrieval*

How could a hierarchical relationship like c-command be used online in language comprehension? One possibility is to differentiate hierarchically available, c-commanding antecedents from distractors at retrieval, through cues targeting hierarchically informed features of items in memory (Lewis and Vasishth, 2005; Lewis et al., 2006). However, as discussed in the Introduction, it is challenging to do this in a cue-based system (Alcocer and Phillips, 2012; Kush, 2013; Kush et al., 2015). This is because the feature content of a single item may not easily include relational information between items as in *x c-commands y* or hierarchical positional information *x is above y in the relevant hierarchy* (e.g., Alcocer and Phillips, 2012; Dillon, 2011; Kush, 2013; Wagers, 2008).

To address this challenge, it has been proposed that item representations may be enriched with features that *approximate* hierarchical relations, rather than directly encode them (Dillon et al., 2014; Kush, 2013; Kush et al., 2015). One such implementation was proposed by Kush (2013): C-commanding items are predictively marked as available based on the evolving phrase structure constructed by the parser. In particular, noun phrases that are either directly attached to the spine of the currently processed clause (i.e., branches immediately dominated by the clause node in the syntactic tree) or serve as its arguments are assigned the feature LOCAL:1, while other items receive LOCAL:0. This feature is dynamically updated during parsing such that, at the onset of a new clause, all LOCAL:1 values from the previous clause are rewritten as LOCAL:0. Although the LOCAL feature does not explicitly encode c-command in memory representations, it allows the parser to approximate the relevant relational constraints of local anaphors (for extended discussion and concerns about the generalizability of this feature, see Kush, 2013). This model maintains a strictly cue-based position on memory retrieval, but reconciles this with the hierarchical nature of language comprehension by dynamically managing feature content so that all and only the hierarchically accessible items in memory at a given point have the relevant LOCAL feature.

Standard cue-based retrieval models posit that cues are combined in an equally weighted, linear fashion at retrieval, but cues could also be combined in a differentially weighted fashion, with hierarchical ones being prioritized over others. This would maximize the reactivation of the target while minimizing the distractor's (e.g., Kush and Phillips, 2014; Parker and Phillips, 2017; Parker, 2019; Yadav et al., 2022) (for a stronger view, see Dillon et al., 2013; Sturt, 2003; Van Dyke and McElree, 2011). Both accounts predict interference from feature-matching distractors, with this effect predicted to be substantially smaller in the differentially weighted cue combination account than in the linear account. Given that we saw relatively little interference, our findings are more compatible with a retrieval model in which hierarchical cues are weighted more heavily than others, ensuring early and efficient access to the target antecedent of the anaphor. In such a model, distractors may cause interference, but this may be too subtle to be detected in medium-sized experiments (Jäger et al., 2015a, 2020), possibly accounting for the inconsistent evidence for interference in our study.

### *6.2. Hierarchical information in memory representations*

Alternatively, hierarchical information may directly shape how linguistic items are organized and maintained in short-term or working memory. On this view, hierarchically available (i.e., c-commanding) items are rapidly accessed not because of retrieval cues targeting them, but because of how they are stored and represented in memory.

This proposal is in line with recent structurally oriented accounts of sentence processing, which posit that linguistic items (e.g., words) are maintained through transient associations with their syntactic positions, with retrieval guided by cues linked to these positions (Cho et al., 2017, 2020; Keshev et al., 2025). These item-to-position associations may also serve a broader organizational function, enabling memory systems to narrow the set of accessible items and improve efficiency for cognitive operations.

On this view, task-relevant, hierarchically prominent items such as locally c-commanding subjects and indirect objects are stored in temporarily active, dynamically updated, privileged memory addresses that facilitate rapid access to them in retrieval. For example, there may be a spatially structured memory state that temporarily binds content (e.g., lexical items) to context (e.g., structural template for the c-command domain), yielding a limited set of elements immediately available for direct access through their associations with current context information, without requiring a search in memory or the postulation of a distinct 'working memory' store (e.g., Foraker and McElree, 2011; McElree, 2006) (see Oberauer, 2002, 2009, for a similar component in his general-purpose working memory architecture, i.e., *the region of direct access*). When c-commanding items are the target of a cognitive operation, i.e., the antecedent of a dependent anaphor, they could be rapidly accessed thanks to their privileged association with this structured region of memory. This account differs from cue-based retrieval models as it attributes rapid access to targets not to their reactivation due to matching features, but to their temporary representational association with privileged structural roles, which results in direct access to them. Consequently, this view does not predict distractors to be reactivated or be accessible and interfere during early stages of retrieval, as they lack this representational advantage. However, the activation of feature-matching distractors may still accumulate over time, potentially leading to increased attention or interference in later stages of processing, as observed in some studies (e.g., Kush and Phillips, 2014; Sturt, 2003).

While both retrieval- and representation-based accounts can capture the rapid access to target antecedents within the anaphor window, disentangling these two requires further investigation, such as computational simulations to model the time-course of access to critical referents and the amount of activation each referent accrues over time (see Nicenboim and Vasishth, 2018). The lack of consistent evidence for the reactivation of feature-matching, plural distractors within this window may favor a privileged representation for hierarchically available items. However, we caution against drawing this conclusion at the moment as it is also possible that we have not been able to observe reactivation as it was too brief or transient to be detected. We leave this for future work.

It is important to note that neither of the accounts discussed above offers a mechanism for directly encoding an abstract, relational representation, such as a direct encoding of the c-command relation between two relata. Instead, the availability of c-commanding items is modeled by virtue of their hierarchically prominent position, implemented either through anticipatory feature assignment or through a reserved, structurally privileged store. However, it is also possible that the rapid use of relational information in syntactic processing uses truly relational representations, the precise nature of which remains to be explored (for relational representations in cognition, see Hafri and Firestone, 2021; Hafri and Trueswell, 2025; Halford et al., 2010).

### *6.3. Subject bias*

While our findings indicate that comprehenders immediately reactivate both c-commanding subjects and indirect objects upon processing the Turkish reciprocal, we see a consistent bias for the subject, suggesting a role for argument role/thematic status in accessing the antecedent above and beyond c-command. This is evident in the higher rate of subject choices in the offline tasks and increased attention to the subject over the indirect object in the visual world studies. The increased availability of subjects may reflect a more prominent ranking as possible antecedents for anaphoric binding (Jackendoff, 1972; Pollard and Sag, 1994). Additionally, the early increase in the visual attention to subjects may be, at least partly, due to the crosslinguistically observed agent preference in sentence comprehension (Schlesewsky and Bornkessel, 2004) and more broadly in cognition (Isasi-Isasmendi et al., 2023; Leslie, 1995). That is, comprehenders often have a tendency to identify and track agents as they interpret an event (e.g., Cohn and Paczynski, 2013; Galazka and Nyström, 2016; Isasi-Isasmendi et al., 2023; Webb et al., 2010).

However, other factors related to the experimental stimuli and methodology may have contributed to the subject bias. For example, we did not control for the salience of subject and indirect object interpretations of the anaphor. Thus, the subject bias could reflect the semantic content of the sentences. The observed bias may also have been boosted by the discourse context: The passive format of the questions may have increased the saliency of the embedded subject relative to the indirect object. Importantly, the availability of indirect objects observed in Experiments 2 and 3 and the lack of a subject bias at the onset of the anaphor window (see Figures 6 and 9 above) show that this effect can be overcome. Lastly, the relative frequency of trials featuring the subject as the only available antecedent may have contributed to this bias. As can be seen in Figures 6 and 9 the subject bias appears to be more pronounced in Experiment 3 compared to Experiment 2. This may be because the ratio of experimental trials with a subject interpretation only relative to those with a subject and an indirect object interpretation for the anaphor was higher in Experiment 3 (16 vs. 8 trials) than in Experiment 2 (12 vs. 18 trials); This could have contributed to within-experiment statistical learning that could have exaggerated the subject bias. Future studies that better control for these factors could reveal the extent of a true underlying subject bias.

## 7. Conclusion

We examined the role of item-to-item, hierarchical relations between noun phrases within a clause as in *x c-commands y* in antecedent retrieval using the local reciprocal in Turkish. In three visual world studies, we measured the moment-by-moment reactivation of c-commanding subject and indirect object antecedents, contrasted with non-c-commanding distractors. The stimuli were carefully constructed to distinguish the role of hierarchical relations from item-specific structural information (i.e., clause, case and subjecthood) and linear order/recency. C-commanding targets (i.e., subjects and indirect objects) were immediately distinguished from distractors upon encountering the reciprocal.

Our findings provide evidence that hierarchical relational information guides antecedent retrieval above and beyond other sources of structural information and linear order at the earliest stages of processing a local anaphor. The immediate availability of c-commanding items can be captured within cue-based activation models in which item representations are equipped with hierarchically informed features that are cued in retrieval (Kush, 2013). Alternatively, this finding may be modeled with memory architectures that directly encode priority for hierarchically prominent items during retrieval.

## Ethics approval

All experimental procedures were approved by the UMass Institutional Review Board (Protocol No.3977; approved on December 7, 2022).

## Credit authorship contribution statement

**Özge Bakay:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Project administration, Writing – original draft, Writing – review & editing, **Faruk Akkuş:** Conceptualization, Formal analysis, Methodology, Project administration, Supervision, Visualization, Writing – review & editing, **Brian Dillon:** Conceptualization, Formal analysis, Methodology, Project administration, Supervision, Visualization, Writing – review & editing.

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| | Reciprocal | | | Verb | | |
|---|---|---|---|---|---|---|
| | Estimate | Std.error | $t/z$ | Estimate | Std.error | $t/z$ |
| *Corrected log subject vs. IO/distractor ratios* | | | | | | |
| *Crossed model* | | | | | | |
| (Intercept) | **0.25** | **0.13** | **2.01** | **0.71** | **0.14** | **5.20** |
| C-command | **-0.51** | **0.23** | **-2.19** | **-0.40** | **0.20** | **-2.001** |
| Match | -0.13 | 0.21 | -0.63 | -0.22 | 0.20 | -1.09 |
| C-command:Match | -0.17 | 0.43 | -0.41 | -0.48 | 0.40 | -1.18 |
| *Nested model* | | | | | | |
| (Intercept) | **0.25** | **0.13** | **2.01** | **0.71** | **0.14** | **5.16** |
| C-command | **-0.51** | **0.23** | **-2.19** | **-0.41** | **0.20** | **-2.02** |
| IO: Match | -0.22 | 0.30 | -0.73 | -0.47 | 0.33 | -1.40 |
| Distractor: Match | -0.05 | 0.30 | -0.16 | -0.02 | 0.28 | 0.07 |
| *New fixations towards subject* | | | | | | |
| *Crossed model* | | | | | | |
| (Intercept) | -0.08 | 0.08 | -0.89 | -0.16 | 0.08 | -1.91 |
| C-command | -0.16 | 0.15 | -1.04 | 0.17 | 0.17 | 0.99 |
| Match | -0.17 | 0.16 | -1.09 | -0.06 | 0.15 | -0.40 |
| C-command:Match | -0.27 | 0.34 | -0.80 | -0.45 | 0.34 | -1.33 |
| *Nested model* | | | | | | |
| (Intercept) | -0.08 | 0.08 | -0.93 | -0.15 | 0.08 | -1.84 |
| C-command | -0.16 | 0.15 | -1.04 | 0.18 | 0.18 | 1.02 |
| IO: Match | -0.30 | 0.21 | -1.44 | -0.32 | 0.25 | -1.27 |
| Distractor: Match | -0.04 | 0.23 | -0.18 | 0.16 | 0.21 | 0.74 |
| *New fixations towards IO/distractor* | | | | | | |
| *Crossed model* | | | | | | |
| (Intercept) | **-0.54** | **0.09** | **-6.18** | **-0.73** | **0.11** | **-6.80** |
| C-command | 0.29 | 0.17 | 1.68 | 0.11 | 0.17 | 0.64 |
| Match | **0.42** | **0.15** | **2.83** | 0.10 | 0.17 | 0.58 |
| C-command:Match | **0.77** | **0.32** | **2.41** | -0.23 | 0.35 | -0.66 |
| *Nested model* | | | | | | |
| (Intercept) | **-0.54** | **0.09** | **-6.22** | **-0.73** | **0.11** | **-6.84** |
| C-command | 0.29 | 0.17 | 1.69 | 0.08 | 0.18 | 0.46 |
| IO: Match | **0.80** | **0.20** | **3.92** | -0.04 | 0.29 | -0.15 |
| Distractor: Match | 0.03 | 0.21 | 0.13 | 0.20 | 0.22 | 0.92 |

Table 6: Coefficient estimates, and corresponding standard errors and $t/z$ values for the fixed effects in mixed-effects linear/logistic regression models on corrected log embedded subject vs. IO/distractor ratios and new fixations towards embedded subject, IO or distractor in Experiment 2. $|T/z|$-values that are bigger than 2 indicate statistical significance, and are given in bold.

| | Reciprocal | | | Verb | | |
|---|---|---|---|---|---|---|
| | Estimate | Std.error | *t/z* | Estimate | Std.error | *t/z* |
| *Corrected log subject vs. IO/distractor ratios* | | | | | | |
| (Intercept) | **0.84** | **0.12** | **6.91** | **1.42** | **0.14** | **10.34** |
| C-command | **-0.80** | **0.16** | **-5.08** | **-0.70** | **0.17** | **-4.17** |
| Case | -0.003 | 0.19 | -0.01 | 0.11 | 0.19 | 0.61 |
| *New fixations towards subject* | | | | | | |
| (Intercept) | **0.24** | **0.11** | **2.19** | -0.11 | 0.08 | -1.31 |
| C-command | **-0.28** | **0.11** | **-2.65** | 0.11 | 0.11 | 0.93 |
| Case | -0.03 | 0.12 | -0.28 | 0.11 | 0.13 | 0.87 |
| *New fixations towards IO/distractor* | | | | | | |
| (Intercept) | **-0.77** | **0.07** | **-11.60** | **-0.41** | **0.06** | **-7.04** |
| C-command | **0.33** | **0.13** | **2.60** | 0.05 | 0.13 | 0.42 |
| Case | 0.02 | 0.13 | 0.16 | -0.05 | 0.13 | -0.38 |

Table 9: Coefficient estimates, and corresponding standard errors and $t/z$ values for the fixed effects in mixed-effects linear/logistic regression models on corrected log embedded subject vs. IO/distractor ratios and new fixations towards the subject, IO or distractor in Experiment 3. $|T/z|$-values that are bigger than 2 indicate statistical significance, and are given in bold.

# Appendix A. A review of studies on local anaphor processing

| 1. Citation | 2. Language | 3. N | 4. Embedding of target or distractor | 5. C-commanding distractor? | 6. Clause-mate distractor? | 7. Case matching? distractor? | 8. Argument role of distractor | 9. Recency of target | 10. Matching feature of distractor | 11. Position of anaphor | 12. Target match effects? | 13. Target vs. distractor bias | 14. Distractor match main effects? | 15-16. Distractor match effect in grammatical sentences? | ungrammatical sentences? |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Event-related potentials* | | | | | | | | | | | | | | | |
| Xiang et al. (2009) (reflexives) | English | 34 | RC | No | No | Yes | Subject | Distant | Gender | Postverbal | Yes | - | - | - | No |
| *Eye-tracking-while-reading* | | | | | | | | | | | | | | | |
| Andrews (2021) E2 (reflexives) | English | 64 | PP | No | Yes | No | Possessor | Distant | Number | Postverbal | cri,+1 (L) | - | No | N/R | No |
| Cunnings and Felser (2013) E1 (low WM) | English | 16 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | cri (E, L); cri,+1 (L) | - | cri+2 (E) | No | No |
| Cunnings and Felser (2013) E1 (high WM) | English | 16 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | No | - | cri+2 (E) | No | No |
| Cunnings and Felser (2013) E2 (low WM) | English | 16 | RC | No | No | Yes | Subject | Distant | Gender | Postverbal | cri+1 (L) | - | No | cri (FAC, E) | No |
| Cunnings and Felser (2013) E2 (high WM) | English | 16 | RC | No | No | Yes | Subject | Distant | Gender | Postverbal | cri+1 (L) | - | No | No | No |
| Cunnings and Sturt (2014) E1 | English | 28 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | cri,+1 (L) | - | No | N/R | N/R |
| Dillon et al. (2013) E1 (reflexives) | English | 40 | RC | No | No | No | Object | Distant | Number | Postverbal | cri (E,L) | - | No | No | No |
| Dillon et al. (2013) E2 | English | 32 | RC | No | No | No | Object | Distant | Number | Postverbal | cri,+1 (L) | - | No | N/R | N/R |
| Felser et al. (2009) E2b (native, c-command) | English | 37 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | - | - | - | No | - |
| Felser et al. (2009) E2b (native, no c-command) | English | 37 | SC | No | No | No | Nonsubject | Recent | Gender | Postverbal | - | - | - | No | - |
| Jäger et al. (2015b) E1 | Chinese | 150 | AC | No | No | - | Subject | Distant | Animacy | Postverbal | cri (E,L); cri-1,+1 (L) | - | cri-1 (L); cri,+1 (E, L) | No | cri-1 (INH, L); cri (INH, E, L); cri+1 (INH, E) |
| Jäger et al. (2015a) E2 | German | 151 | RC | No | No | Yes | Subject | Distant | Gender | Preverbal | - | - | - | No | - |
| Jäger et al. (2015a) E3 (reflexives) | Swedish | 35 | RC | No | No | Yes | Subject | Distant | Gender | Postverbal | - | - | - | No | - |
| Jäger et al. (2020) (reflexives) | English | 190 | RC | No | No | No | Object | Distant | Number | Postverbal | N/R | - | N/R | cri (INH, E) | cri (FAC, L) |
| King et al. (2012) E2 | English | 48 | RC | No | No | No | Object | Distant | Gender | Postverbal | cri (E, L); cri+1 (E) | - | cri (E,L) | N/R | N/R |
| Parker and Phillips (2017) E1 (1-feature-mismatch) | English | 30 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | cri,-1 (L) | - | No | N/R | No |
| Parker and Phillips (2017) E1 (2-feature-mismatch) | English | 30 | SC | Yes | No | Yes | Subject | Recent | Gender & Animacy | Postverbal | cri (L) | - | No | N/R | cri (FAC, L) |
| Parker and Phillips (2017) E2 (reflexives) | English | 30 | RC | No | No | Yes | Subject | Distant | Number & Animacy | Postverbal | - | - | - | - | cri (FAC, E, L); cri+1 (L) |
| Parker and Phillips (2017) E3 (1-feature-mismatch) | English | 24 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | cri (L) | - | No | N/R | No |
| Parker and Phillips (2017) E3 (2-feature-mismatch) | English | 24 | SC | Yes | No | Yes | Subject | Recent | Gender & Number | Postverbal | cri,-1 (L) | - | No | N/R | cri (FAC, E, L) |
| Patil et al. (2016) | English | 40 | RC | No | No | Yes | Subject | Distant | Gender | Postverbal | cri (L) | - | N/R | cri (INH, E) | No |
| Sezer (2020) E3 | Turkish | 40 | RC | No | No | No | Subject | Recent | Animacy | Preverbal | cri+1: (E); cri,+1,+2,+3 (L) | - | cri,+1,+2 (L) | cri,+2 (INH, L) | cri+1 (INH, L) |
| Sezer (2020) E4 | Turkish | 30 | AC | No | No | Yes | Subject | Distant | Animacy | Preverbal | cri,+1,+2 (L); cri+3 (E) | - | cri (E, L) | cri (INH, L) | cri (INH, E) |
| Slogett (2017) E1b (speech verbs) | English | 37 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | cri,+1 (L) | - | No | No | cri (FAC, E, L) |
| Slogett (2017) E1b (perception verbs) | English | 37 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | cri,+1 (L) | - | No | No | cri+1 (FAC, E) |
| Slogett (2017) E2 | English | 45 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | cri,+1 (L) | - | cri,+1 (L) | No | cri,+1 (FAC, L) |
| Sturt (2003) E1 | English | 24 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | cri (E,L); cri+1,+2 (L) | - | cri+2 (L) | cri+2 (FAC, L) | No |
| Sturt (2003) E2 | English | 24 | RC | No | No | No | Object | Distant | Gender | Postverbal | cri (E,L); cri+1 (L) | - | No | N/R | N/R |
| *Eye-tracking-while-listening* | | | | | | | | | | | | | | | |
| Clackson et al. (2011) E2 (control, reflexives) | English | 40 | AC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | - | Yes | - | No | - |
| Clackson and Heyer (2014) | English | 42 | AC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | - | Yes | - | Yes | - |
| Han et al. (2021) E2 (reflexives) | English | 28 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | - | Yes | - | No | - |
| Runner and Head (2014) (reflexives, SRC) | English | 25 | RC | No | No | No | Object | Distant | Gender | Postverbal | - | Yes | Yes | Yes | - |
| Runner and Head (2014) (reflexives, ORC) | English | 25 | RC | No | No | Yes | Subject | Distant | Gender | Postverbal | - | Yes | Yes | No | - |
| *Priming* | | | | | | | | | | | | | | | |
| Nicol (1988) E1 (reflexives) | English | 61 | SC | Yes | No | Yes/No | Subject/Object | Recent | Gender | Postverbal | - | Yes | - | No | - |
| *Self-paced reading* | | | | | | | | | | | | | | | |
| Badecker and Straub (2002) E3 (reflexives) | English | 32 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | - | - | - | cri+2 (INH) | - |
| Badecker and Straub (2002) E4 | English (recip) | 36 | SC | Yes | No | Yes | Subject | Recent | Number | Postverbal | - | - | - | cri+1-4 combined (INH) | - |
| Badecker and Straub (2002) E5 (reflexives) | English | 28 | NP | No | Yes | No | Possessor | Recent | Gender | Postverbal | - | - | - | No | - |
| Badecker and Straub (2002) E6 | English | 48 | SC | No | No | No | Nonsubject | Recent | Gender | Postverbal | - | - | - | No | - |
| Chen et al. (2012) | Chinese | 120 | AC | No | No | - | Subject | Distant | Animacy | Postverbal | - | - | - | cri,+1 (INH) | - |
| Clifton et al. (1999) E1 (reflexives) | English | 48 | PP | No | Yes | No | Possessor/Adjunct | Distant | Number | Postverbal | - | - | - | No | - |
| Clifton et al. (1999) E2 (reflexives) | English | 48 | PP | No | Yes | No | Possessor/Adjunct | Distant | Number & Gender | Postverbal | - | - | - | No | - |
| Clifton et al. (1999) E3 (reflexives) | English | 49 | SC | Yes | No | Yes | Subject | Recent | Gender | Postverbal | - | - | - | No | - |
| Jäger et al. (2015a) E1 | German | 144 | RC | No | No | Yes | Subject | Distant | Gender | Preverbal | - | - | - | No | - |
| Keshev and Meltzer-Asscher (2024) E1 | Hebrew | 78 | PP | No | Yes | No | Possessor | Distant | Gender | Postverbal | - | - | - | - | cri (INH), cri+1 (FAC) |
| King et al. (2012) E1 | English | 48 | RC | No | No | No | Object | Distant | Number | Postverbal | cri+1 | - | No | N/R | N/R |
| Kush and Phillips (2014) | Hindi (recip) | 32 | RC | No | No | No | Object | Distant | Number | Preverbal | cri+1 | - | No | No | No |
| Laurinavichyute et al. (2017) E1 (reflexives) | German | 111 | RC | No | No | Yes | Subject | Distant | Gender | Preverbal | - | - | - | No | - |
| Laurinavichyute et al. (2017) E2A | Russian | 109 | RC | No | No | Yes | Subject | Distant | Gender | Preverbal | - | - | - | cri-1,+2 (INH) | - |
| Laurinavichyute et al. (2017) E2B | Russian | 112 | RC | No | No | Yes | Subject | Distant | Gender | Postverbal | - | - | - | cri-3 (INH) | - |
| Pizarro-Guevara and Dillon (2021) | Tagalog | 80 | RC | No | No | Yes | Subject | Distant | Number | Postverbal | cri | - | cri | N/R | N/R |
| Yon (2019) E8 | Korean | 24 | AC | No | No | Yes/No | Subject | Distant | Animacy | Preverbal | - | - | - | cri,+1 (INH) | - |

Table A.10: A review of the studies on the real-time processing of local anaphors, grouped by the methodology employed. The review excludes studies that do not have structurally unavailable distractors (e.g., Arslan et al., 2025; Gračanin-Yuksek et al., 2017), or those with anaphors that are analyzed as exempt from standard binding constraints such as those in picture noun phrases (e.g., Runner et al., 2003). Unless stated as *recip(rocal)* under Column 2, all studies were on reflexives. Column 3 shows the number of participants. Columns 4-11 detail experimental stimuli. Column 4 indicates the hierarchical position of the more deeply embedded antecedent, i.e., target or distractor. Column 5 refers to whether the distractor c-commands the anaphor. Columns 6-7 and 10 refer to the match between the target and the distractor in terms of clausemateness, (abstract) case marking and features, respectively. Column 8 indicates the argument role of the distractor. Column 9 indicates the linear position of the target relative to the distractor, defining both of their positions with respect to the anaphor. Columns 12-16 summarize empirical findings. Column 12 reports differences between conditions with a feature-matching vs. -mismatching target. Column 13 reports whether there was a bias towards the target over the distractor in visual world or priming studies. Columns 14-16 reports differences between conditions with feature-matching vs. -mismatching distractor. For Columns 12-16

, 'Yes' means a significant effect, 'No' means a non-significant effect, 'N/R' means the test was not run or reported, and '-' that the test does not apply. Statistical inferences follow main analyses originally reported as significant, and excludes exploratory analyses or marginal effects. Some studies are split by measure or stimuli, and thus appear on multiple lines. AC = adverbial clause, cri = critical region with anaphor, cri-/+ = region before or after the critical region as indexed with the following number, E = early reading measures in eye-tracking, FAC = facilitatory effects (i.e., speed-up with a feature-matching distractor), INH = inhibitory effects (i.e., slow-down with a feature-matching distractor), L = late reading measures in eye-tracking, NP = noun phrase, SC = sentential complement, PP = pre/post-positional phrase, RC = relative clause, WM = working memory.

To aid in interpreting the table above, we summarize Experiment 2 from Andrews (2021) with reference to the table columns, as an example. The experiment employed the eye-tracking-while-reading methodology to investigate the processing of English reflexives (Columns 1-2) in postverbal position (Column 11), recruiting 64 native speakers of English (Column 2). The stimuli had a 2x2 design crossing grammaticality (i.e., number (mis)match between the target and the reflexive) and distractor number (i.e., (mis)match between the target and the distractor, Column 10), as shown in (11). The target antecedent was the head of a subject NP, while the distractor was a possessor noun embedded in a PP in the subject phrase (Columns 4 & 8). The distractor was in the same clause as the target (Column 6), but it did not match with the target in terms of (abstract) case marking (i.e., genitive vs. nominative) (Column 7) nor did it c-command the reflexive (Column 5). The target antecedent was always the linearly more distant antecedent from the reflexive compared to the distractor (Column 9).

(11) a. The aunts of the actress(es) definitely embarrassed themselves at the gala before.
b. *The aunt of the actress(es) definitely embarrassed themselves at the gala before.

The results showed evidence for a grammaticality effect in the critical reflexive and its spillover region in late reading measures (Column 12). There was no evidence for an overall effect of distractor number (Column 14), nor was there an effect of distractor number in ungrammatical sentences (Column 16). Statistical tests for the effect of distractor number in grammatical sentences were not reported (Column 15).

## Appendix B. Offline norming tasks for Experiments 1-3

### *Appendix B.1. Experiment 1: Antecedent selection task*

Prior to Experiment 1, an offline antecedent selection task ($N_{participant}$ = 32, $M_{age}$ = 22.8) verified the c-command constraint in the experimental stimuli. The task was conducted online on PCIbexFarm (Zehr and Schwarz, 2018). Each sentence was presented on the same screen along with a question asking about the referent(s) of the anaphor and four referent options, i.e., target, distractor, matrix subject and 'others'. The order of the referent options was randomized across items. Participants were instructed to click on the referent(s) that answered the question. They were specifically instructed to click on all possible options, if more than one answer was possible. There were 36 fillers, including those probing the availability of multiple referents as correct answers. Each participant received 80 Turkish liras (∼\$4.5 at the time of the study) for compensation.

Participants' ($N = 31$; 1 excluded due to low accuracy) responses are summarized in Table B.11. Participants chose the target antecedent above 98% of the time across the experiment, confirming that the c-commanding, plural subject was a possible antecedent for *birbirleri*. The non-c-commanding distractor was overall chosen only around 14% of the time.

| | Target % | Distractor % | Matrix Subject % | Others % |
|---|---|---|---|---|
| Recent Target, Distractor Match | 97.8 (1.3) | 15.2 (4.9) | 2.2 (1.7) | 4.8 (1.8) |
| Recent Target, Distractor Mismatch | 100 | 10.8 (4.4) | 5.4 (3.5) | 3.2 (2.2) |
| Distant Target, Distractor Match | 97.8 (1) | 19.9 (4.9) | 4.3 (1.7) | 5.6 (2.9) |
| Distant Target, Distractor Mismatch | 98.9 (0.4) | 11.3 (4.4) | 5.8 (2.5) | 5.9 (3.6) |

Table B.11: Mean percentages (and SEs over by-participant means) for choices in the antecedent selection task for Experiment 1.

A mixed-effects logistic regression model was fit to the choice data for the distractor to systematically examine the distractor choices. The results revealed a main effect of MATCH, with more choices of a plural than a singular distractor ($z = 2.83$). A post-hoc comparison showed that there were more choices of a plural than a singular distractor when the target was distant ($z = 2.83$), but choices of a singular and plural distractor were comparable when the target was recent ($z = 1.76$). Overall, the task revealed some amount of distractor choices when it was plural and linearly proximate to the anaphor, although the overall rate of these responses is quite low.

*Appendix B.2. Experiment 2: Norming task for the c-command relation of the IO*

The indirect object (IO) of a ditransitive verb in Turkish may originate in a position that is higher or lower than the direct object (DO) (Öztürk, 2005). Whereas the IO c-commands the DO in its base position in the former case, it doesn't in the latter. To confirm that our stimuli had the IO c-command the anaphor in the DO position in its base position, we conducted a norming task with various ditransitive verbs. Twenty-four native speakers of Turkish ($M_{age} = 24.6$) participated in this task and received 100 Turkish liras (~\$4 at the time of the study) for compensation. Their task was to decide on the availability of IO>DO and DO>IO ordering of 16 ditransitive verbs when a bound-variable reading between a quantified IO and a lower, possessed DO is enforced as in *Ayşe asked every$_i$ director about their$_i$ cameraman* vs. *Ayşe asked about their$_i$ cameraman to every$_i$ director*. For twelve verbs, participants chose the IO>DO ordering over DO>IO in more than 70% of cases, suggesting that the IO c-commands the DO in the configurations formed with these verbs. Seven of these verbs that can be optionally used as a transitive or ditransitive verb were used as the embedded verb in the experimental stimuli of Experiments 2-3.

*Appendix B.3. Experiment 2: Antecedent selection task*

An offline antecedent selection task ($N_{participant} = 42$, $M_{age} = 25.2$) was conducted to verify the c-command constraint in the experimental stimuli in Experiment 2. There were 42 fillers, including those probing the availability of multiple referents as well as those that required a dative-marked noun as the correct answer. The rest of the procedure was the same as in the previous antecedent selection task. Each participant received 100 Turkish liras (~\$4 at the time of the study) as compensation.

Participants' ($N = 39$; 3 excluded due to low accuracy) choices of antecedents are summarized in Table B.12. The results showed that overall, participants chose the embedded subject above 91% of the time. The IO was chosen around 35%, and the distractor was chosen around 27% of the time across the experiment.

A mixed-effects logistic regression model fit to the choice data for IO/Distractor in the 2x2 design, i.e., the four conditions with a plural subject, showed a main effect of the number of dative nouns, with more choices of plural than singular dative nouns ($z = 3.70$). There was no main effect of the c-command of dative nouns or an interaction between c-command and number of dative nouns ($z$'s $\leq 1.13$). A post-hoc comparison revealed that plural IOs were chosen more often than singular ones ($z = 2.91$), but the choices for singular and plural distractors were comparable ($z = 1.74$). A $t$-test comparing the choices of the singular subject and plural IO in the control condition showed significantly more choices of the subject ($t = 13.98$).

| | Embedded Subject % | IO/ Distractor % | Matrix Subject % | Others % |
|---|---|---|---|---|
| Plural Subject, IO Match | 93.2 (1.5) | 33.8 (4.9) | 2.1 (1.4) | 3.4 (1.4) |
| Plural Subject, IO Mismatch | 96.6 (1.4) | 20.9 (4.4) | 1.7 (1.3) | 4.3 (1.8) |
| Plural Subject, Distractor Match | 98.7 (0.7) | 29.9 (5.3) | 0.9 (0.6) | 2.9 (1.8) |
| Plural Subject, Distractor Mismatch | 97 (1.2) | 23.9 (5) | 0.4 (0.4) | 3.9 (2.3) |
| Singular Subject, IO Match | 71.8 (3.8) | 49.6 (4.6) | 4.7 (1.7) | 11.1 (3.5) |

Table B.12: Mean percentages (and SEs over by-participant means) for choices in the antecedent selection task for Experiment 2.

We found that plural IOs were more frequently selected as antecedents than singular IOs. However, singular IOs were still chosen approximately 20% of the time along with the plural subject in the same sentence chosen around 96% of the time. Second, the overall selection rate of IOs was much smaller than subjects. Third, in the control condition, although the plural IO was chosen more often than in the other conditions, the singular subject was still chosen more often than the IO.

Lastly, the overall selection rate of distractors was unexpectedly high, but feature-matching distractors were not reliably more available than mismatching ones. One explanation for the unexpectedly high selection of distractors could lie in the postposition used in the stimulu, i.e., *göre* ‘according to’. This postposition may take wide scope over the entire sentence (e.g., Ernst, 2001) and, in doing so, enable binding in the absence of a c-command relation. To test whether the distractor is less available when participants are limited to a single choice, we conducted a forced-choice task using the same stimuli (see below).

*Appendix B.4. Experiment 2: Forced-choice antecedent selection task*

To address the issue of high selection rate of distractors in the antecedent selection task (see above), we put the same materials from Experiment 2 into a forced choice antecedent selection task ($N_{participant} = 35$, $M_{age} = 27.5$). The list of experimental and filler items was the same as in the antecedent selection task above. The procedure of this task was similar to the visual world study. That is, the participants were first given the question only, and then on a new screen the sentence along with the four referent options: embedded subject, IO/distractor, matrix subject and ‘others’. They were instructed to click on the most likely referent that answered the question. Each participant received 120 Turkish liras (∼$4 at the time of the study) for compensation.

Participants’ ($N = 30$; 5 excluded due to low accuracy) choices of antecedents are summarized in Table B.13. Results demonstrated that overall participants chose the embedded subject around 85% of the time. The IO was chosen around 20%, and the distractor around 2% of the time across the experiment.

| | Embedded Subject % | IO/ Distractor % | Matrix Subject % | Others % |
|---|---|---|---|---|
| Plural Subject, IO Match | 80 (3.8) | 17.8 (3.7) | 0 | 2.2 (1.3) |
| Plural Subject, IO Mismatch | 91.7 (2.5) | 7.8 (2.5) | 0 | 0.6 (0.6) |
| Plural Subject, Distractor Match | 97.2 (1.2) | 2.2 (1.1) | 0 | 0.6 (0.6) |
| Plural Subject, Distractor Mismatch | 95.6 (1.8) | 2.2 (1.3) | 0 | 2.2 (1.3) |
| Singular Subject, IO Match | 61.1 (6.4) | 33.9 (6) | 0 | 5 (2.3) |

Table B.13: Mean percentages (and SEs over by-participant means) for choices in the forced-choice antecedent selection task for Experiment 2.

A mixed-effects logistic regression model fit to the choice data for IO/Distractor in the 2x2 design showed a main effect of c-command, with more choices of IO than distractor ($z = 4.54$). A post-hoc comparison showed that the plural IO was chosen more often than the singular IO ($z = 2.44$), but the choices for singular and plural distractors were comparable ($z = 0.11$). A $t$-test comparing the choices of the subject and IO in the control condition showed more choices of the subject ($t = 9.58$).

Overall, the forced-choice task showed that distractor selections were negligible when participants were limited to a single antecedent choice, in line with our speculation that the higher distractor selection rate above may stem from participants' efforts to accommodate a wide-scope reading of the postposition *göre* 'according to' when multiple selections were allowed. Given that local binding is preferred over binding via wide scope in the forced-choice task, we used the stimuli above for Experiment 2, but address this further by using a different set of postpositions in Experiment 3 (see below). Lastly, singular subjects were chosen more often than plural IOs in the control condition, and singular IOs were selected over plural subjects approximately 8% of the time; suggesting that c-command and subjecthood may play a more dominant role in antecedent selection than number alone.

*Appendix B.5. Experiment 3: Antecedent selection task*

An offline antecedent selection task ($N_{participant} = 30$, $M_{age} = 25.7$) verified the c-command constraint in the experimental stimuli in Experiment 3. There were 46 fillers, including those probing the availability of multiple referents. The rest of the procedure was the same as in previous antecedent selection tasks. Each participant received 120 Turkish liras (~\$4 at the time of the study) for compensation.

Participants' ($N = 25$; 5 excluded due to low accuracy) choices of antecedents are in Table B.14. Results showed that the embedded subject was chosen above

| | Embedded Subject % | IO/ Distractor % | Matrix Subject % | Others % |
|---|---|---|---|---|
| IO, Dative | 92 (2) | 46 (7) | 4.5 (4) | 4 (2.3) |
| Distractor, Dative | 99 (0.7) | 12.5 (3.7) | 4.5 (3.5) | 3.9 (2.2) |
| Distractor, Genitive | 95.5 (1.9) | 20 (5.9) | 6 (3.6) | 6.5 (3) |

Table B.14: Mean percentages (and SEs over by-participant means) for choices in the antecedent selection task for Experiment 3.

95% of the time; the c-commanding IO was chosen around 46% of the time; and the distractor around 16% of the time across the experiment.

A mixed-effects logistic regression model fit to the choice data for IO/distractor revealed a main effect of c-command, with more choices of IO than distractor ($z$ = 5.17). A comparison within the two Distractor conditions showed that the genitive-marked distractor was chosen more often than the dative-marked distractor ($z$ = 2.45). The increased choices of the genitive-marked distractors may suggest that participants occasionally considered these distractors as potential antecedents due to their case similarity to c-commanding subjects. Regardless, the choice data confirmed the unavailability of distractors in binding the reciprocal.

## Appendix C. Visual and auditory stimuli in Experiments 1-3

All visual stimuli used in Experiments 1–3 were downloaded from Adobe Stock photos and were drawn by the same artist in a consistent style and format. Prior to the visual world studies, the stimuli were normed for codability. The participants ($N = 20$, $M_{age} = 22.7$) were asked to type the occupation of the character in each image in an online survey. Each participant received 80 Turkish liras (~$4.5 at the time of the study) for compensation. Seventy-four of the pictures that were given the same label by at least 75% of the participants were used in the experiments.

The auditory stimuli in Experiments 1-3 were recorded using the GarageBand application on a Macbook Air by the first author. Each experimental sentence was recorded in three parts and then merged into a single audio file: (i) the matrix subject, (ii) the segment containing the target and distractor phrases (which varied by condition), and (iii) the final segment containing the adverb, anaphor, and two verbs. The same audio files were used for the first and third segments across all conditions of an item. To ensure that the critical anaphor onset was aligned across conditions, the three segments were merged using fixed onset points for each item. If the second segment was shorter in a condition, silent padding was added to its end so that the third segment began at a consistent time.

The three-part recording method introduced two prosodic boundaries, which presumably enhanced naturalness by inducing pauses at appropriate points. A pause following the matrix subject is expected in Turkish, as subjects are typically default topics and optionally followed by a pause (Göksel and Kerslake, 2005; Kamali, 2008). A pause before the adverb is also plausible, as the preceding segment contains three words, well within the typical prosodic phrase length observed in Turkish speakers who pause at every 4.2 words on average (range: 2.9–7.8) during read-aloud story tasks (according to Deniz's (2014) analysis of Nash's (1973) data).

Before data collection, the critical time points for the onsets of the reciprocal and verb windows in each item set were established by the first author via examination of spectrograms in Praat.

Filler items were recorded as single segments but included two prosodic breaks in positions identical to those in the experimental sentences. Lead-in sentences and questions were recorded individually, as a single segment. To normalize for loudness, the average intensity of all recordings was set to 70.00 dB. The set of auditory files for the example stimuli in Experiment 1 (i.e., the stimuli in (5) and (6) in Section 3.1) can be found at the OSF repository.

## Appendix D. Raw gaze proportions for the duration of the experimental sentence in Experiment 1

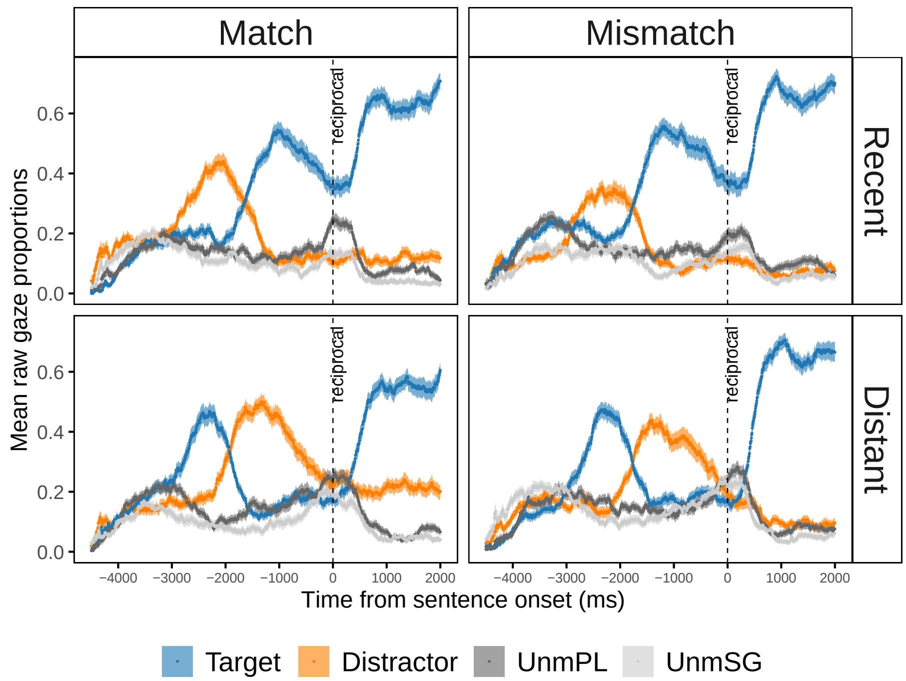


Figure D.12: Mean raw gaze proportions towards the four interest areas in Experiment 1. Error bands show SEs over by-participant means. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Proportions are shown for the duration of the entire sentence, with dashed lines at time point 0 time-locked to the onset of the critical anaphor.

## Appendix E. Model formulas for the statistical analyses in Experiments 1-3

### *Appendix E.1. Model formulas in Experiment 1*

*Gaze proportion analyses*

- Reciprocal, crossed: Target vs. distractor ∼ Recency * Match + (1 + Recency + Match | participant) + (1 + Match + Match:Recency || item), optimizer = 'bobyqa'
- Reciprocal, nested: Target vs. distractor ∼ Recency + Recent: Match + Distant: Match + (1 + Recency + Distant: Match | participant) + (1 + Recent: Match + Distant: Match || item)
- Verb, crossed: Target vs. distractor ∼ Recency * Match + (1 + Recency + Match || participant) + (1 + Recency * Match || item)
- Verb, nested: Target vs. distractor ∼ Recency + Recent: Match + Distant: Match + (1 + Recency + Distant: Match || participant) + (1 + Recency + Distant: Match | item), optimizer = 'bobyqa'

*Analyses on new fixations towards target*

- Reciprocal, crossed: Target ∼ Recency * Match + (1 + Recency * Match || participant) + (0 + Recency | item)
- Reciprocal, nested: Target ∼ Recency + Recent: Match + Distant: Match + (1 + Recency + Recent: Match + Distant: Match || participant) + (0 + Recency + Recent: Match | item), optimizer = 'bobyqa'
- Verb, crossed: Target ∼ Recency * Match + (0 + Match | participant) + (1 + Recency || item)
- Verb, nested: Target ∼ Recency + Recent: Match + Distant: Match + (1 + Recent: Match + Distant: Match || participant) + (1 + Recency || item)

*Analyses on new fixations towards distractor*

- Reciprocal, crossed: Distractor ∼ Recency * Match + (1 + Match | participant) + (1 + Match + Recency:Match || item)

- Reciprocal, nested: Distractor ~ Recency + Recent: Match + Distant: Match + (1 | participant) + (1 | item)
- Verb, crossed: Distractor ~ Recency * Match + (1 + Match || participant) + (1 + Match + Recency:Match || item)
- Verb, nested: Distractor ~ Recency + Recent: Match + Distant: Match + (1 | participant) + (1 + Recent: Match + Distant: Match || item)

*Appendix E.2. Model formulas in Experiment 2*

*Gaze proportion analyses*

- Reciprocal, crossed: Subject vs. IO/distractor ~ C-command * Match + (1 + C-command | participant) + (1 + C-command || item)
- Reciprocal, nested: Subject vs. IO/distractor ~ C-command + IO: Match + Distractor: Match + (1 + C-command | participant) + (1 + C-command || item)
- Verb, crossed: Subject vs. IO/distractor ~ C-command * Match + (1 | participant) + (1 + Match || item)
- Verb, nested: Subject vs. IO/distractor ~ C-command + IO: Match + Distractor: Match + (1 | participant) + (1 + IO: Match | item)

*Analyses on new fixations towards subject*

- Reciprocal, crossed: Subject ~ C-command * Match + (1 + C-command + Match || participant) + (1 + Match + C-command:Match | item)
- Reciprocal, nested: Subject ~ C-command + IO: Match + Distractor: Match + (1 + C-command || participant) + (0 + IO: Match + Distractor: Match | item)
- Verb, crossed: Subject ~ C-command * Match + (1 + C-command * Match || participant) + (0 + C-command:Match | item)
- Verb, nested: Subject ~ C-command + IO: Match + Distractor: Match + (1 + C-command + IO: Match | participant)

*Analyses on new fixations towards IO/distractor*

- Reciprocal, crossed: IO/distractor ∼ C-command * Match + (1 + C-command || participant) + (0 + C-command + C-command:Match || item)
- Reciprocal, nested: IO/distractor ∼ C-command + IO: Match + Distractor: Match + (1 + C-command || participant) + (0 + C-command || item)
- Verb, crossed: IO/distractor ∼ C-command * Match + (1 + C-command | participant) + (1 + Match + C-command:Match || item)
- Verb, nested: IO/distractor ∼ C-command + IO: Match + Distractor: Match + (1 + C-command | participant) + (1 + IO: Match || item)

*Appendix E.3. Model formulas in Experiment 3*

*Gaze proportion analyses*

- Reciprocal: Subject vs. IO/distractor ∼ C-command + Case + (1 + C-command + Case | participant) + (1 + Case || item), optimizer = 'bobyqa'
- Verb: Subject vs. IO/distractor ∼ C-command + Case + (1 + C-command + Case | participant) + (1 + Case || item)

*Analyses on new fixations towards subject*

- Reciprocal: Subject ∼ C-command + Case + (1 | participant) + (1 | item)
- Verb: Subject ∼ C-command + Case + (1 + C-command || participant) + (1 + C-command || item)

*Analyses on new fixations towards IO/distractor*

- Reciprocal: IO/distractor ∼ C-command + Case + (1 + C-command | participant) + (1 | item)
- Verb: IO/distractor ∼ C-command + Case + (0 + C-command | participant) + (1 | item)

## Appendix F. Raw gaze proportions for the duration of the experimental sentence in Experiment 2

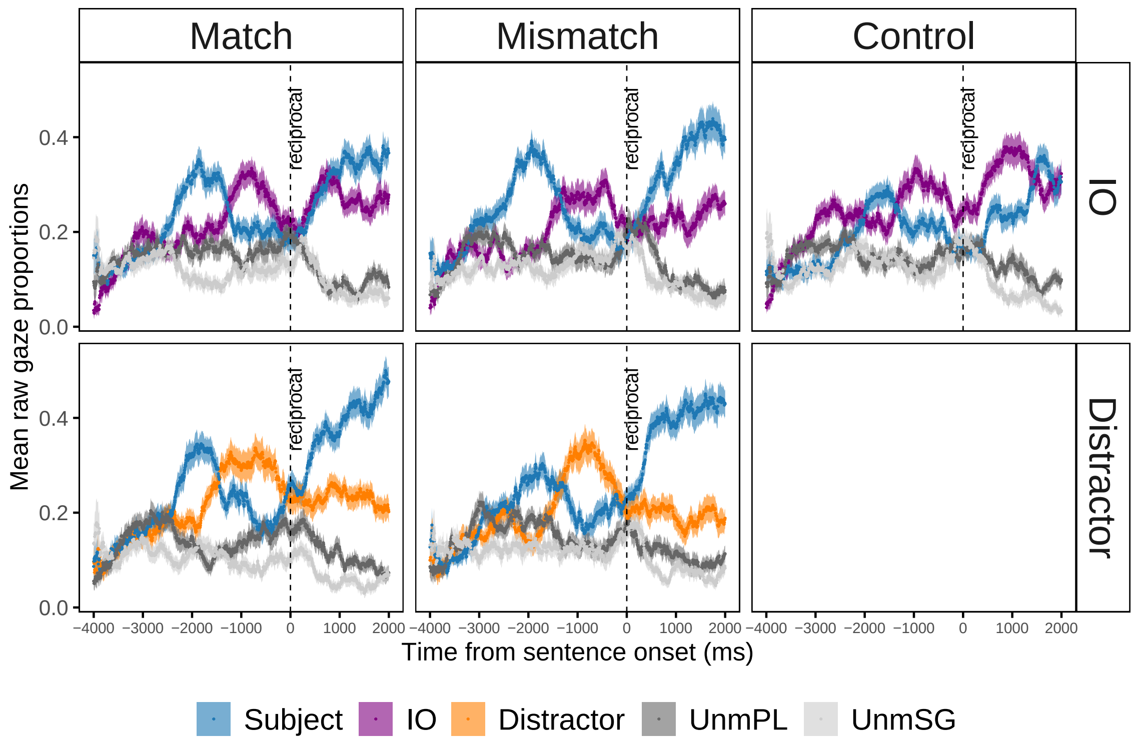


Figure F.13: Mean raw gaze proportions towards the four interest areas in Experiment 2. Error bands show SEs over by-participant means. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Proportions are shown for the duration of the entire sentence, with dashed lines at time point 0 time-locked to the onset of the critical anaphor.

## Appendix G. Power analysis on sample size in Experiment 3

We ran a power analysis based on the data from Experiment 2 on the conditions in (10-a)-(10-b) to estimate the sample size of a high-powered study replicating the difference between plural IO and plural dative-marked distractor. We ran a model that examined new fixations towards the IO vs. distractor on a dataset that only included new fixations towards the subject and IO/distractor. The results showed that there were more new fixations towards the IO than the distractor (*Est.* $= .78$, $SE = .34$, $z = 2.30$). We conducted 1000 simulations based on this model using the SIMR package in R (Green and MacLeod, 2016). Simulations showed that a sample size of 102 subjects would have a power of 82.4% (95% CI: 79.9 - 84.7%) to detect a slightly smaller effect size ($\beta = .70$).

We also ran the same model on a dataset that included all new fixations, i.e., those towards all four regions, as we did in the main analyses. The results of this model also showed more fixations towards the IO than the distractor ($z = 2.66$), but this model had a bigger $|z|$ value than the previous one. In order to be more conservative, we ran the power analysis using the model with a smaller $z$ value.

## Appendix H. Raw gaze proportions for the duration of the experimental sentence in Experiment 3

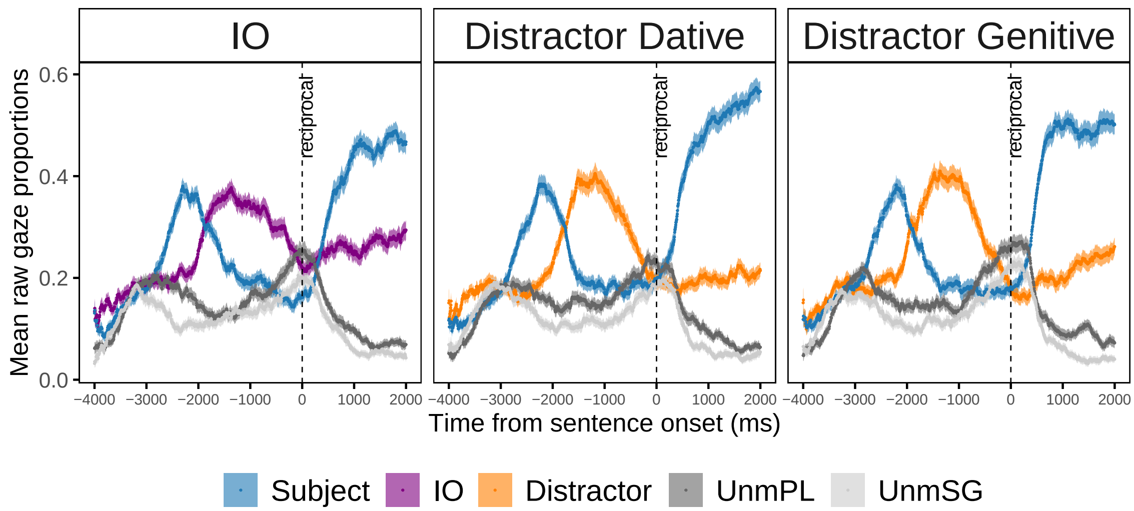


Figure H.14: Mean raw gaze proportions towards the four interest areas in Experiment 3. Error bands show SEs over by-participant means. UnmPL and UnmSG refer to the unmentioned plural and singular referents, respectively. Proportions are shown for the duration of the entire sentence, with dashed lines at time point 0 time-locked to the onset of the critical anaphor.